

# Stage 1 and Stage 2 Results

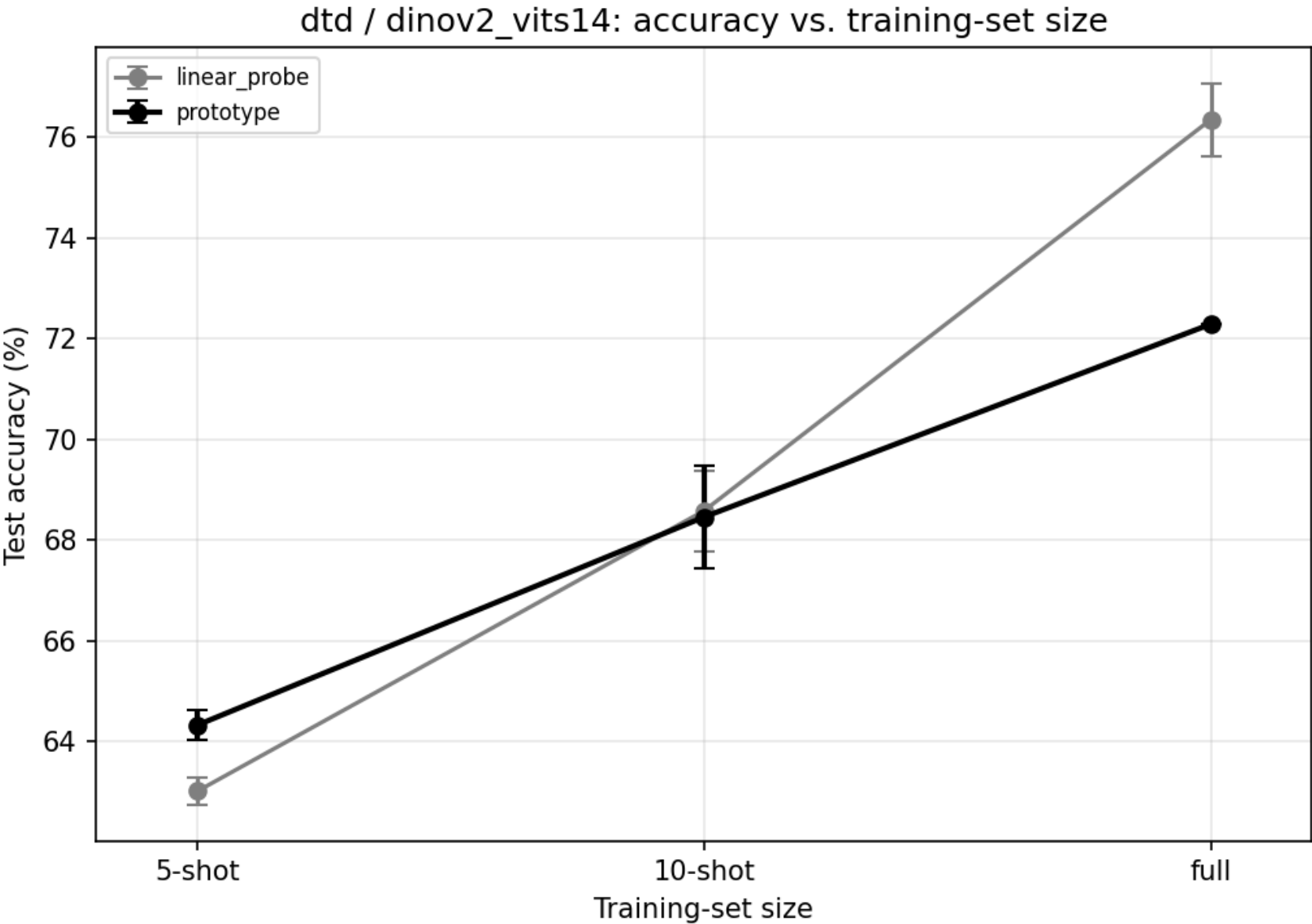
| Dataset | Encoder       | Method        | T  | K-shot | Runs | Test Accuracy    | Delta            |
|---------|---------------|---------------|----|--------|------|------------------|------------------|
| dtd     | dinov2_vits14 | linear_probe  | -  | 5      | 3    | 63.01% +/- 0.27% | -                |
| dtd     | dinov2_vits14 | linear_probe  | -  | 10     | 3    | 68.58% +/- 0.80% | -                |
| dtd     | dinov2_vits14 | linear_probe  | -  | full   | 3    | 76.35% +/- 0.72% | -                |
| dtd     | dinov2_vits14 | fm_cls_rolled | 4  | 10     | 3    | 68.78% +/- 0.61% | +0.20% +/- 0.19% |
| dtd     | dinov2_vits14 | fm_cls_guided | 4  | 10     | 3    | 69.63% +/- 0.37% | +1.05% +/- 0.87% |
| dtd     | dinov2_vits14 | prototype     | -  | 5      | 3    | 64.33% +/- 0.30% | -                |
| dtd     | dinov2_vits14 | prototype     | -  | 10     | 3    | 68.46% +/- 1.01% | -                |
| dtd     | dinov2_vits14 | prototype     | -  | full   | 1    | 72.29%           | -                |
| dtd     | dinov2_vits14 | fm_standard   | 4  | 5      | 3    | 61.31% +/- 0.75% | -3.01% +/- 0.46% |
| dtd     | dinov2_vits14 | fm_standard   | 12 | 5      | 3    | 61.70% +/- 0.83% | -2.62% +/- 0.55% |
| dtd     | dinov2_vits14 | fm_standard   | 4  | 10     | 3    | 65.60% +/- 0.59% | -2.85% +/- 1.45% |
| dtd     | dinov2_vits14 | fm_standard   | 12 | 10     | 3    | 65.92% +/- 0.57% | -2.54% +/- 1.48% |
| dtd     | dinov2_vits14 | fm_standard   | 4  | full   | 1    | 74.63%           | +2.34%           |
| dtd     | dinov2_vits14 | fm_standard   | 12 | full   | 1    | 74.79%           | +2.50%           |
| dtd     | dinov2_vits14 | fm_rolled     | 4  | 5      | 3    | 57.48% +/- 1.32% | -6.84% +/- 1.02% |
| dtd     | dinov2_vits14 | fm_rolled     | 12 | 5      | 3    | 57.50% +/- 1.04% | -6.83% +/- 0.73% |
| dtd     | dinov2_vits14 | fm_rolled     | 4  | 10     | 3    | 61.74% +/- 0.48% | -6.72% +/- 1.46% |
| dtd     | dinov2_vits14 | fm_rolled     | 12 | 10     | 3    | 61.74% +/- 0.27% | -6.72% +/- 1.28% |
| dtd     | dinov2_vits14 | fm_rolled     | 4  | full   | 1    | 73.51%           | +1.22%           |
| dtd     | dinov2_vits14 | fm_rolled     | 12 | full   | 1    | 72.93%           | +0.64%           |
| dtd     | dinov2_vits14 | cls_finetune  | 4  | 10     | 3    | 68.90% +/- 0.80% | +0.32% +/- 0.00% |
| dtd     | dinov2_vits14 | fm_cls_joint  | 4  | 10     | 3    | 69.41% +/- 0.35% | +0.83% +/- 0.46% |
| dtd     | resnet18      | linear_probe  | -  | 5      | 3    | 45.80% +/- 0.75% | -                |
| dtd     | resnet18      | linear_probe  | -  | 10     | 3    | 52.39% +/- 0.74% | -                |
| dtd     | resnet18      | linear_probe  | -  | full   | 3    | 62.55% +/- 0.51% | -                |
| dtd     | resnet18      | prototype     | -  | 5      | 3    | 46.93% +/- 0.32% | -                |

| Dataset    | Encoder  | Method        | T  | K-shot | Runs | Test Accuracy    | Delta             |
|------------|----------|---------------|----|--------|------|------------------|-------------------|
| dtd        | resnet18 | prototype     | -  | 10     | 3    | 51.97% +/- 0.54% | -                 |
| dtd        | resnet18 | prototype     | -  | full   | 1    | 58.78%           | -                 |
| dtd        | resnet18 | fm_standard   | 4  | 5      | 3    | 44.82% +/- 1.45% | -2.11% +/- 1.19%  |
| dtd        | resnet18 | fm_standard   | 12 | 5      | 3    | 45.14% +/- 1.36% | -1.79% +/- 1.07%  |
| dtd        | resnet18 | fm_standard   | 4  | 10     | 3    | 48.65% +/- 1.02% | -3.32% +/- 0.52%  |
| dtd        | resnet18 | fm_standard   | 12 | 10     | 3    | 49.13% +/- 1.17% | -2.84% +/- 0.65%  |
| dtd        | resnet18 | fm_standard   | 4  | full   | 1    | 59.52%           | +0.74%            |
| dtd        | resnet18 | fm_standard   | 12 | full   | 1    | 59.57%           | +0.80%            |
| dtd        | resnet18 | fm_rolled     | 4  | 5      | 3    | 40.51% +/- 0.75% | -6.42% +/- 0.45%  |
| dtd        | resnet18 | fm_rolled     | 12 | 5      | 3    | 40.60% +/- 0.85% | -6.33% +/- 0.53%  |
| dtd        | resnet18 | fm_rolled     | 4  | 10     | 3    | 40.62% +/- 1.00% | -11.35% +/- 1.20% |
| dtd        | resnet18 | fm_rolled     | 12 | 10     | 3    | 41.65% +/- 0.90% | -10.32% +/- 0.88% |
| dtd        | resnet18 | fm_rolled     | 4  | full   | 1    | 56.06%           | -2.71%            |
| dtd        | resnet18 | fm_rolled     | 12 | full   | 1    | 56.01%           | -2.77%            |
| flowers102 | resnet18 | linear_probe  | -  | 5      | 3    | 75.13% +/- 0.22% | -                 |
| flowers102 | resnet18 | linear_probe  | -  | 10     | 3    | 83.22% +/- 0.14% | -                 |
| flowers102 | resnet18 | linear_probe  | -  | full   | 3    | 83.22% +/- 0.14% | -                 |
| flowers102 | resnet18 | fm_cls_rolled | 4  | 10     | 3    | 84.15% +/- 0.06% | +0.93% +/- 0.08%  |
| flowers102 | resnet18 | fm_cls_guided | 4  | 10     | 3    | 84.24% +/- 0.36% | +1.02% +/- 0.34%  |
| flowers102 | resnet18 | prototype     | -  | 5      | 3    | 69.62% +/- 0.32% | -                 |
| flowers102 | resnet18 | prototype     | -  | 10     | 3    | 75.22% +/- 0.00% | -                 |
| flowers102 | resnet18 | prototype     | -  | full   | 1    | 75.22%           | -                 |
| flowers102 | resnet18 | fm_standard   | 4  | 5      | 3    | 69.19% +/- 1.02% | -0.43% +/- 0.85%  |
| flowers102 | resnet18 | fm_standard   | 12 | 5      | 3    | 69.59% +/- 1.01% | -0.02% +/- 0.82%  |
| flowers102 | resnet18 | fm_standard   | 4  | 10     | 3    | 76.83% +/- 0.11% | +1.62% +/- 0.11%  |
| flowers102 | resnet18 | fm_standard   | 12 | 10     | 3    | 77.17% +/- 0.21% | +1.95% +/- 0.21%  |
| flowers102 | resnet18 | fm_standard   | 4  | full   | 1    | 76.79%           | +1.58%            |
| flowers102 | resnet18 | fm_standard   | 12 | full   | 1    | 76.94%           | +1.72%            |

| Dataset    | Encoder  | Method       | T  | K-shot | Runs | Test Accuracy    | Delta            |
|------------|----------|--------------|----|--------|------|------------------|------------------|
| flowers102 | resnet18 | fm_rolled    | 4  | 5      | 3    | 62.54% +/- 1.32% | -7.07% +/- 1.12% |
| flowers102 | resnet18 | fm_rolled    | 12 | 5      | 3    | 63.07% +/- 0.68% | -6.54% +/- 0.40% |
| flowers102 | resnet18 | fm_rolled    | 4  | 10     | 3    | 72.46% +/- 0.38% | -2.76% +/- 0.38% |
| flowers102 | resnet18 | fm_rolled    | 12 | 10     | 3    | 72.65% +/- 0.37% | -2.57% +/- 0.37% |
| flowers102 | resnet18 | fm_rolled    | 4  | full   | 1    | 72.89%           | -2.33%           |
| flowers102 | resnet18 | fm_rolled    | 12 | full   | 1    | 73.07%           | -2.15%           |
| flowers102 | resnet18 | cls_finetime | 4  | 10     | 3    | 83.43% +/- 0.07% | +0.22% +/- 0.17% |
| flowers102 | resnet18 | fm_cls_joint | 4  | 10     | 3    | 84.22% +/- 0.09% | +1.00% +/- 0.14% |

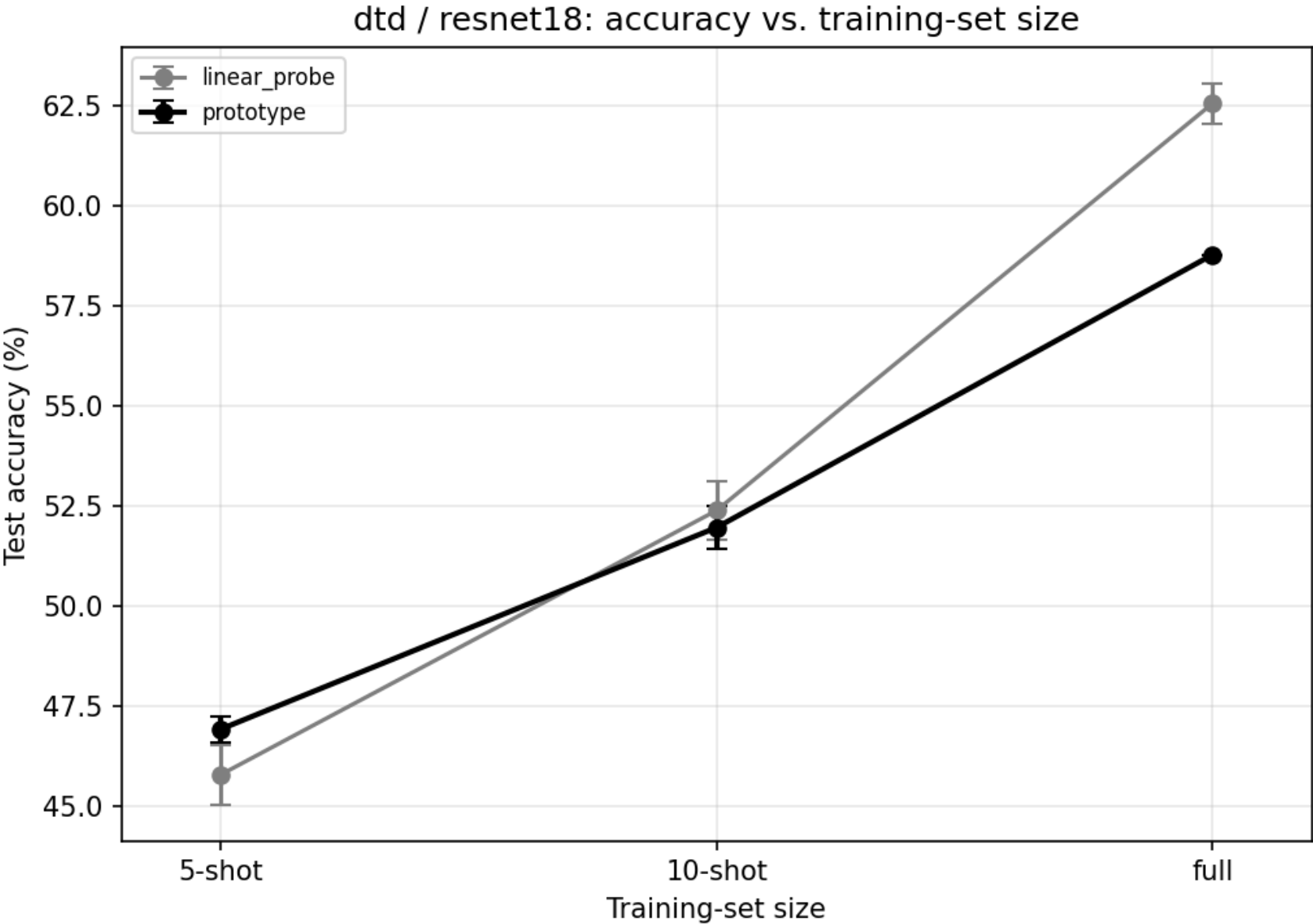
Accuracy vs. training-set size

dtd / dinov2\_vits14



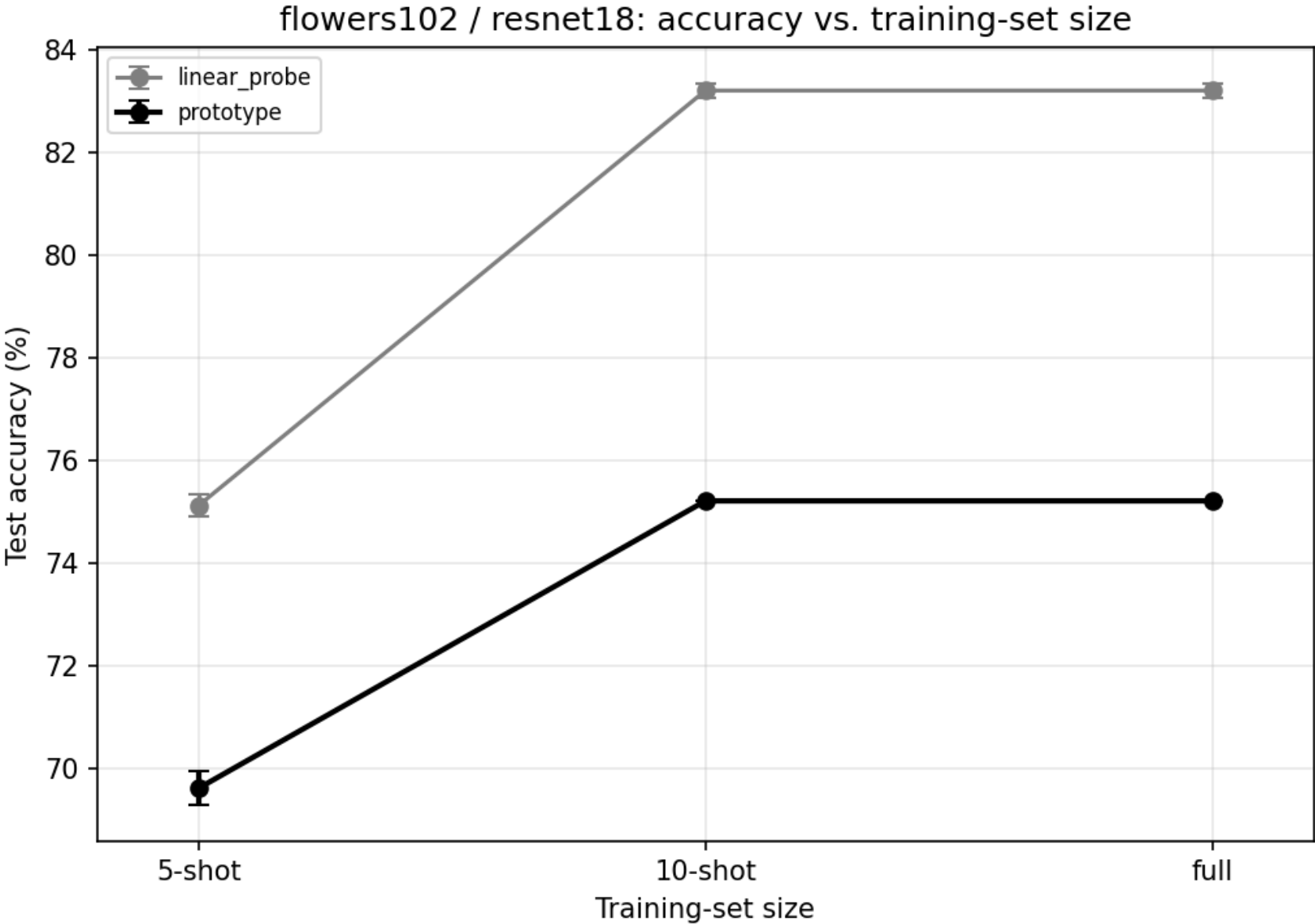
Accuracy vs. training-set size

dtd / resnet18



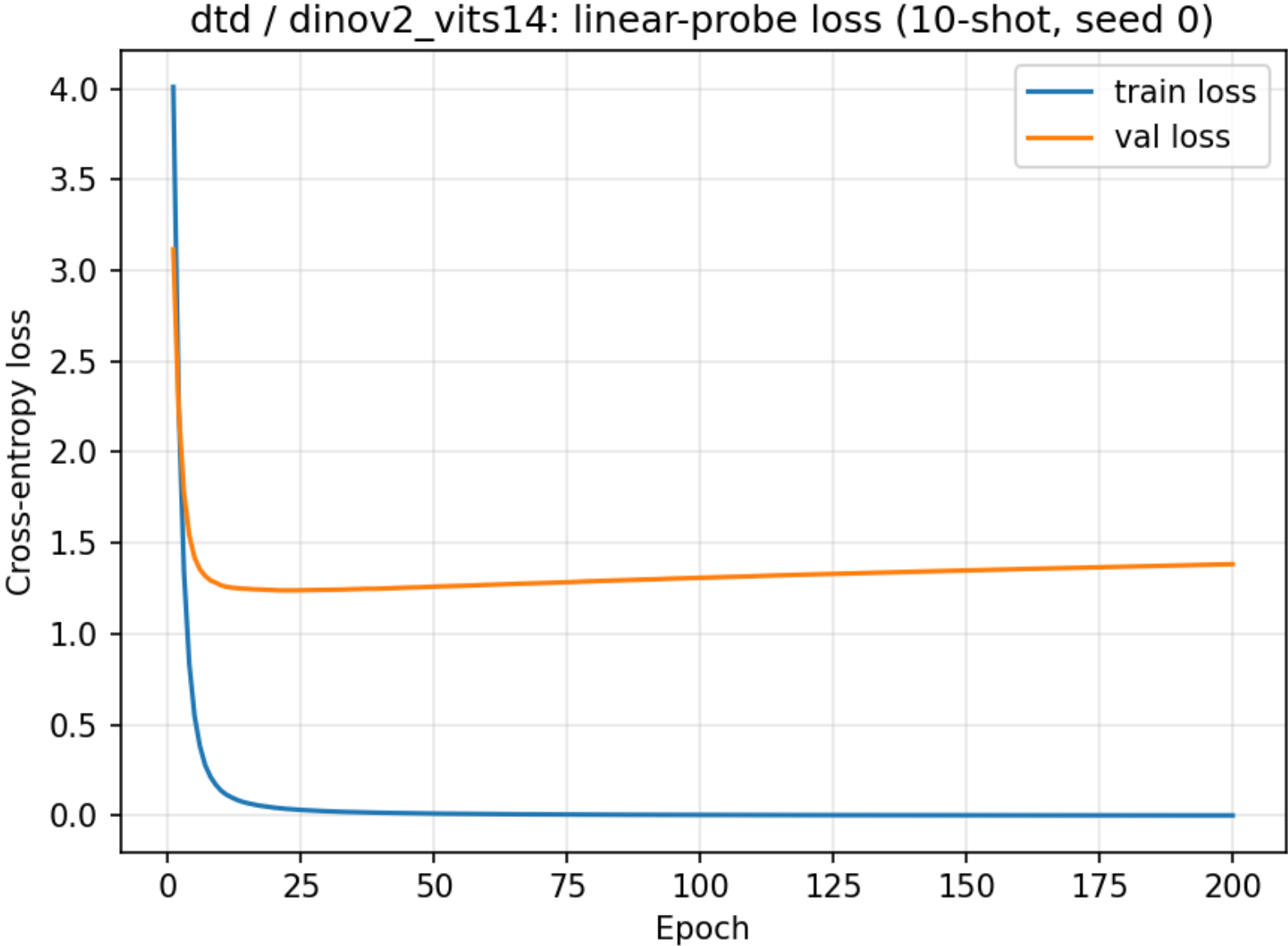
Accuracy vs. training-set size

flowers102 / resnet18



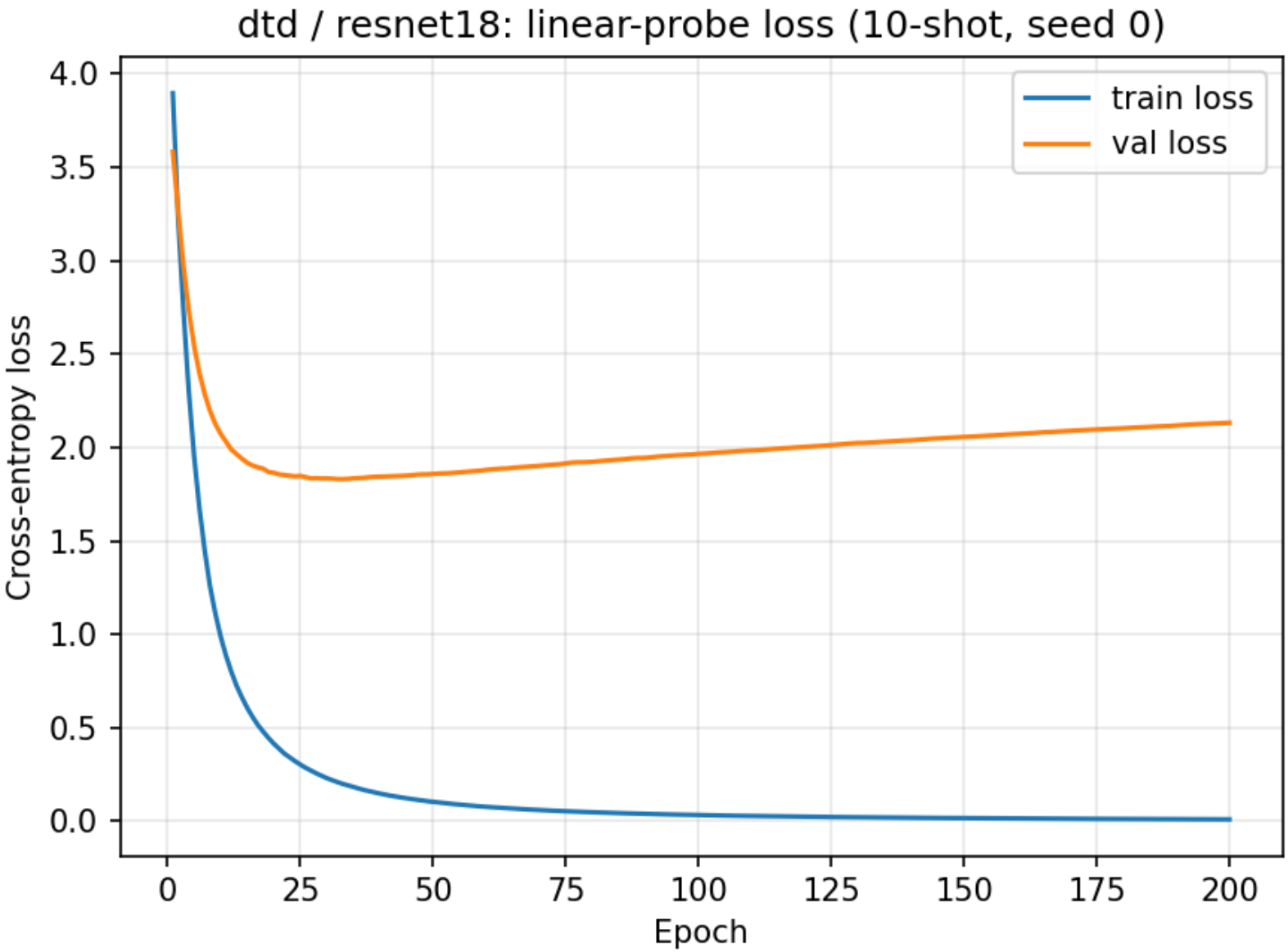
Training / validation loss (10-shot, seed 0)

dtd / dinov2\_vits14



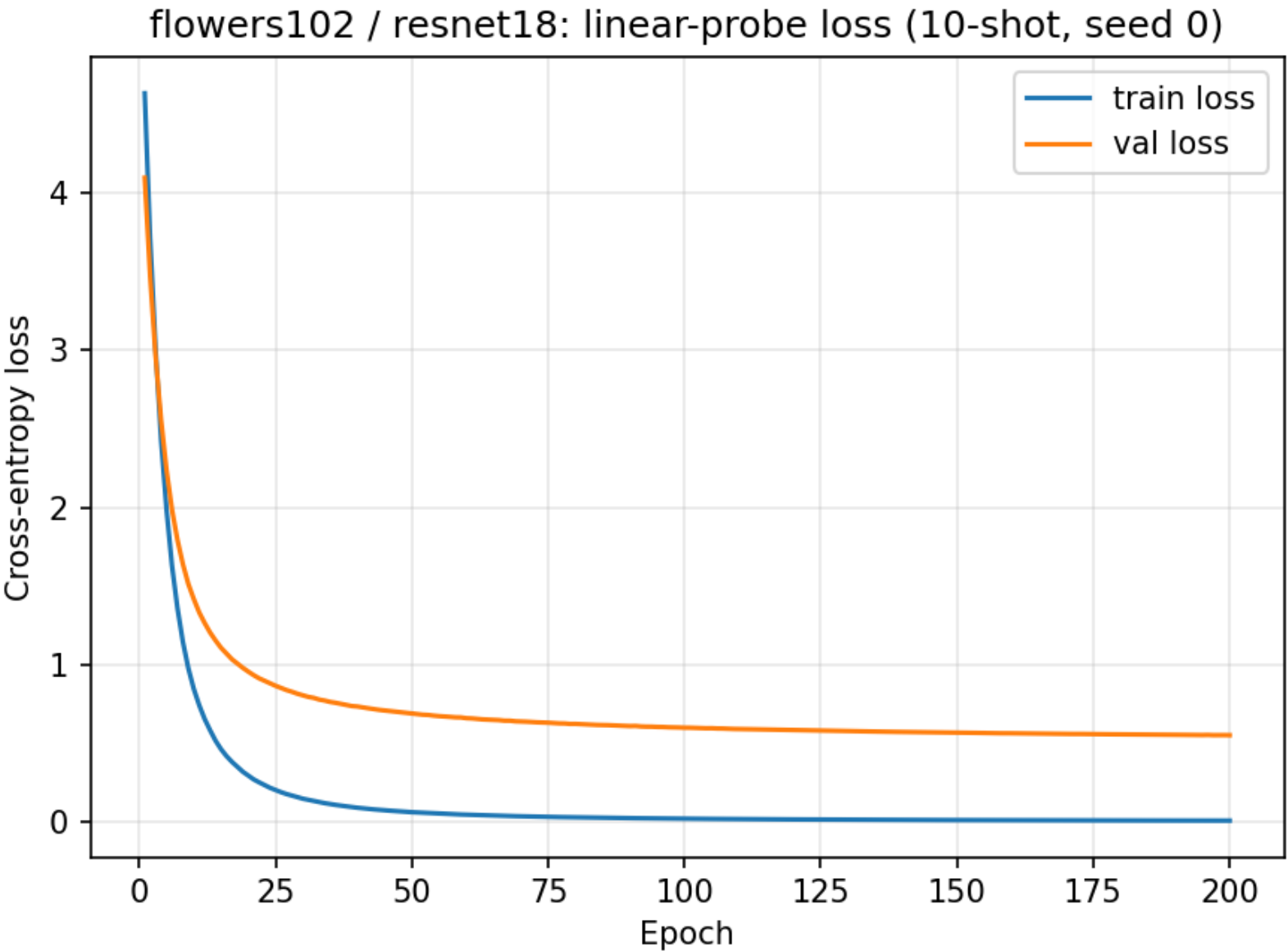
Training / validation loss (10-shot, seed 0)

dtd / resnet18



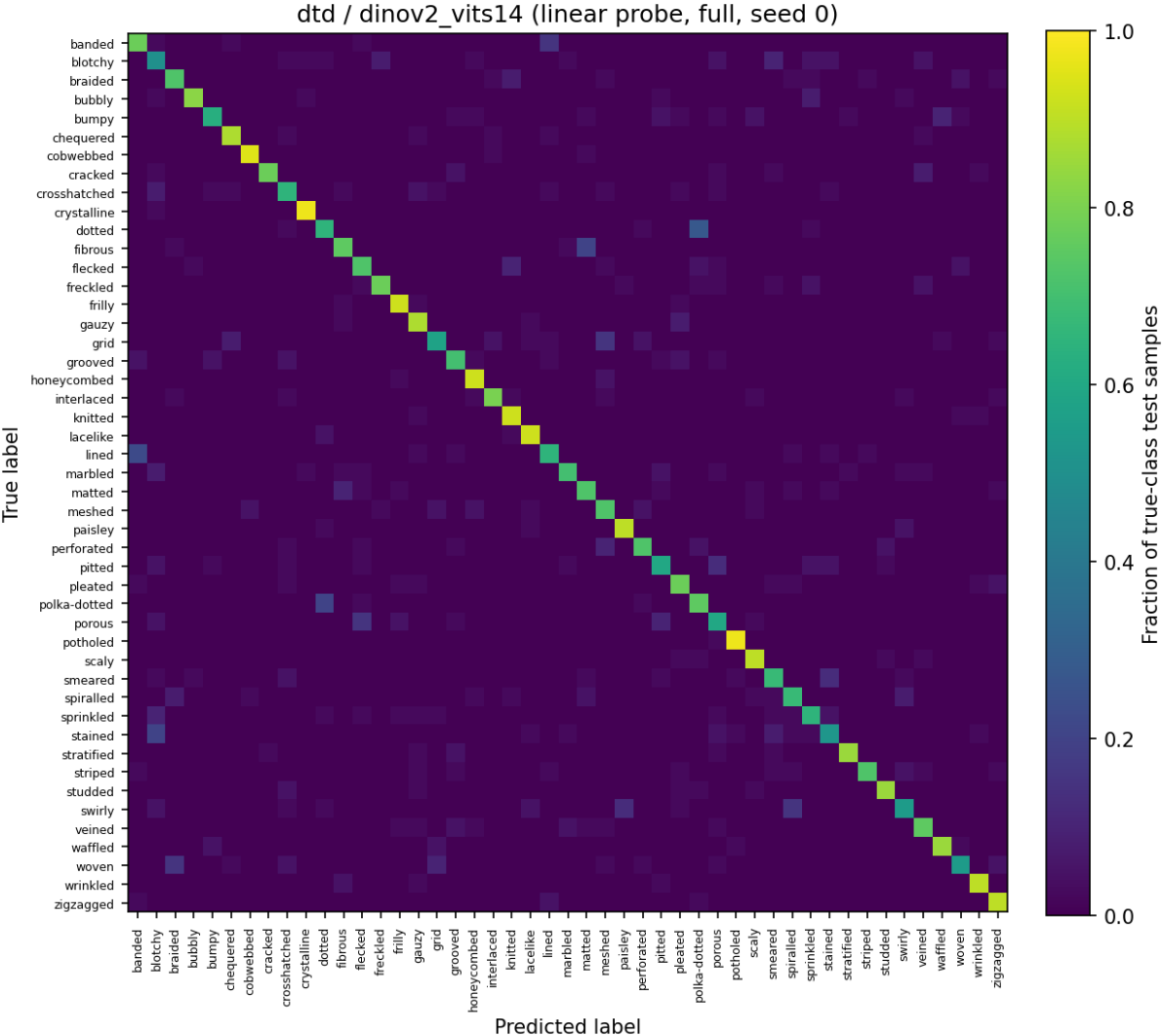
Training / validation loss (10-shot, seed 0)

flowers102 / resnet18



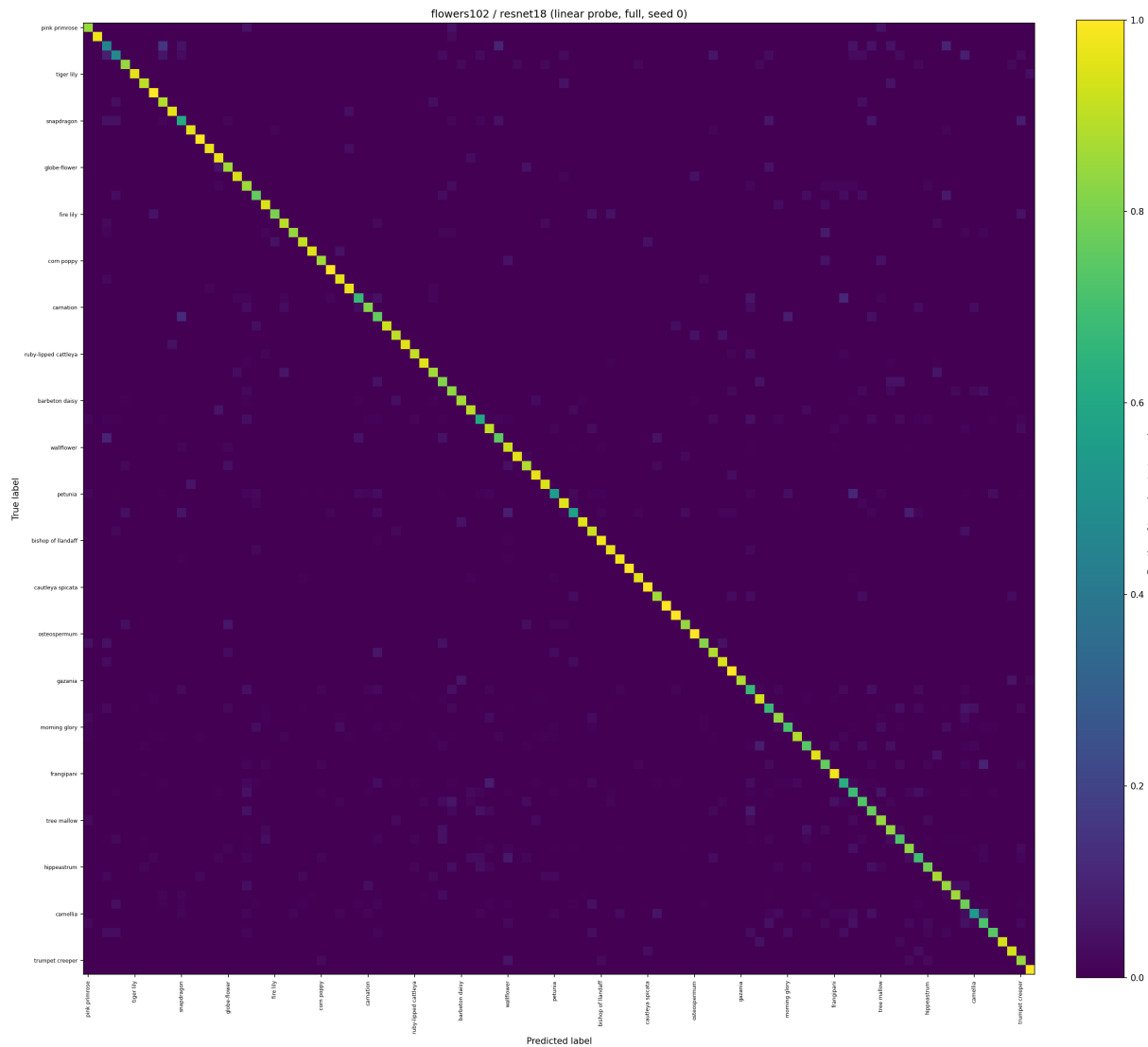
Confusion matrices (row-normalized)

dtd / dinov2\_vits14



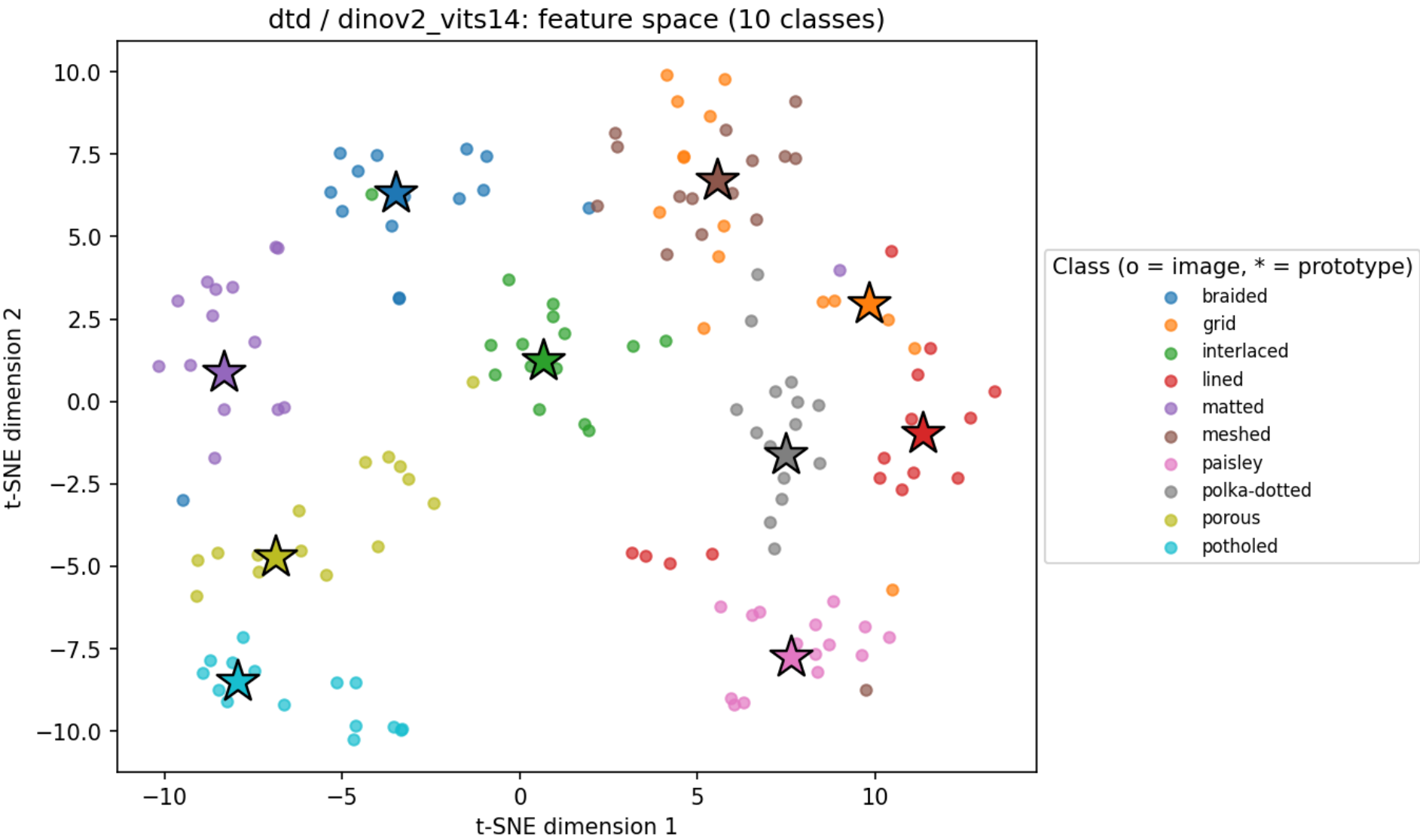
# Confusion matrices (row-normalized)

flowers102 / resnet18



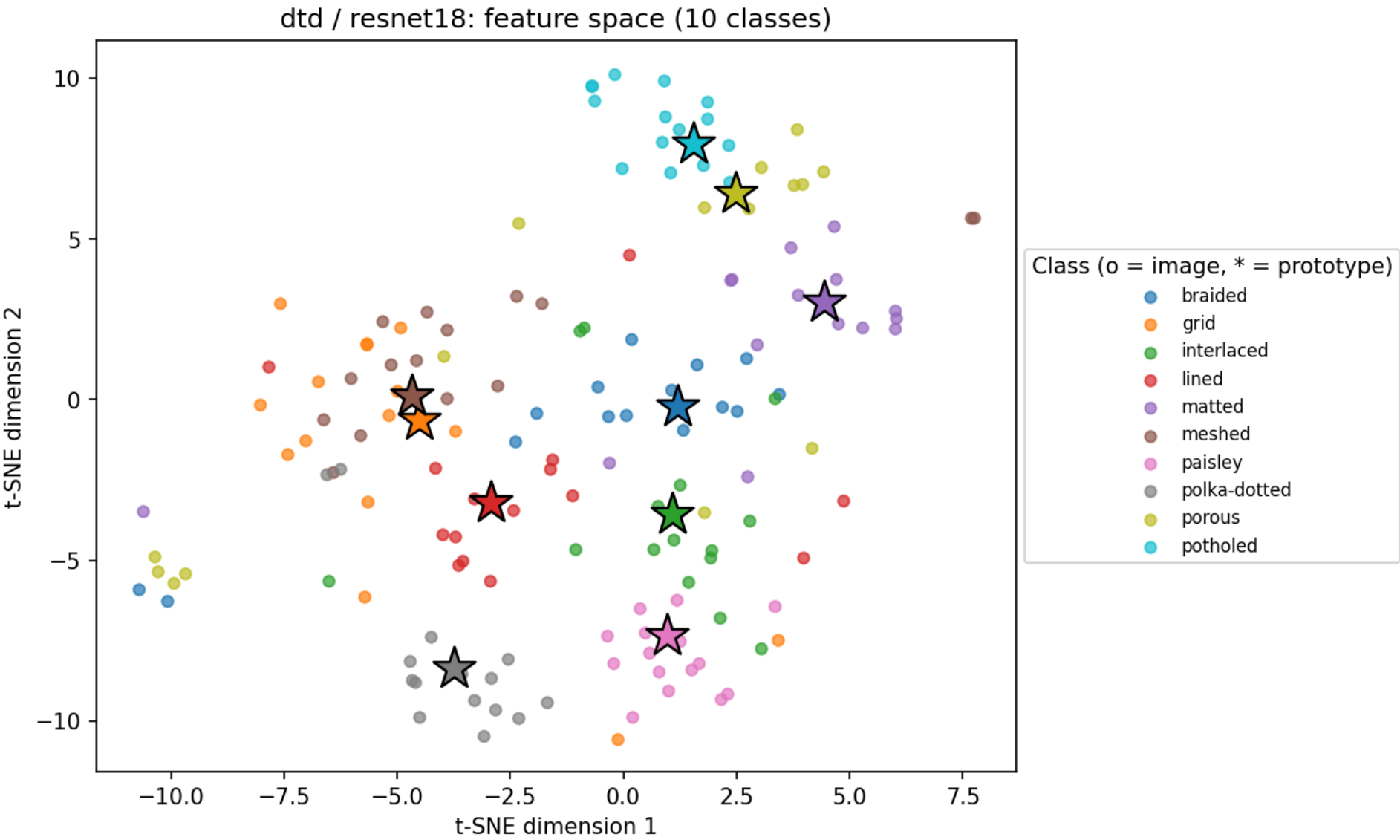
Feature-space visualizations (t-SNE)

dtd / dinov2\_vits14



Feature-space visualizations (t-SNE)

*dtd / resnet18*



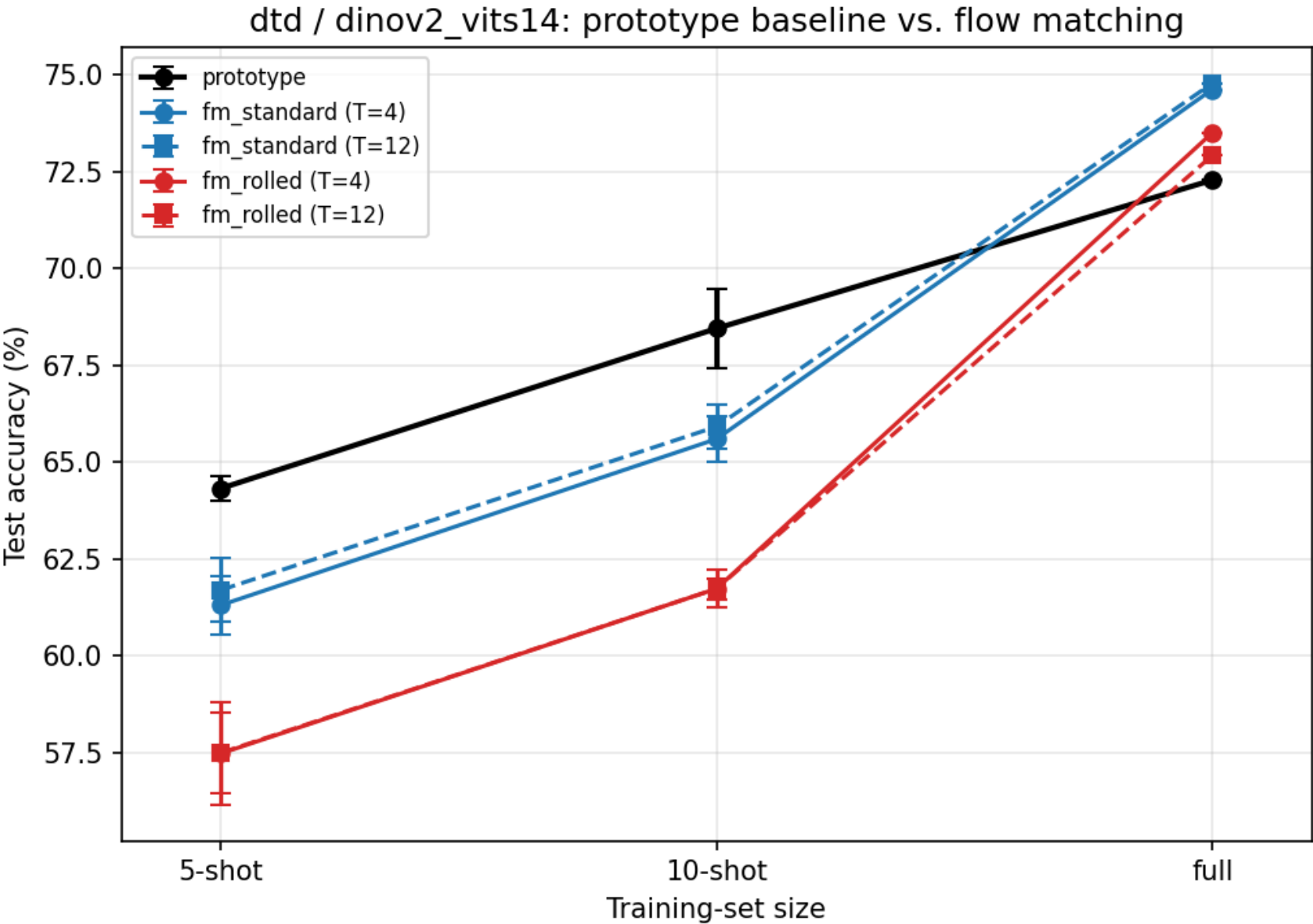
Feature-space visualizations (t-SNE)

flowers102 / resnet18



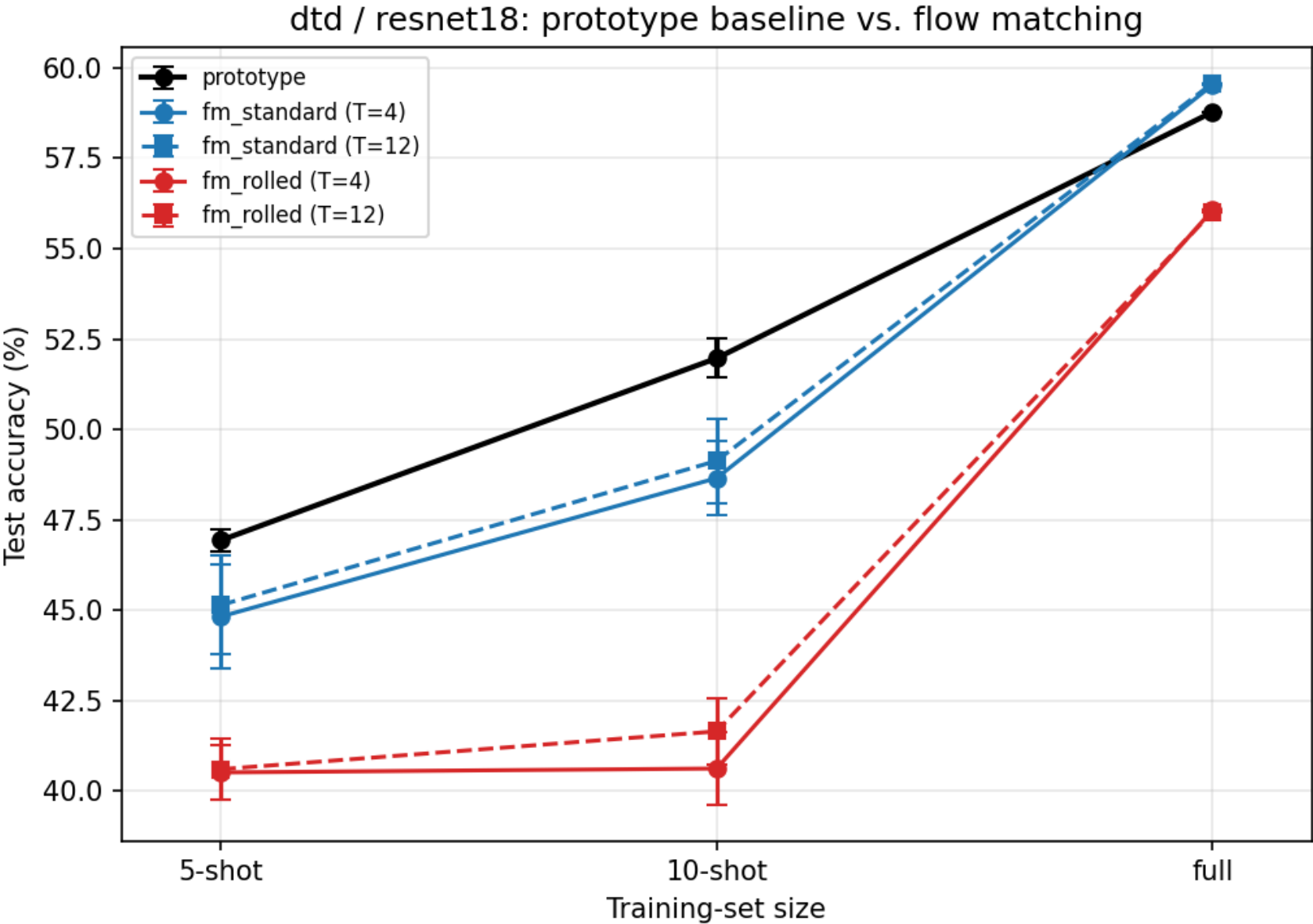
Stage 2: accuracy vs. training-set size

dtd / dinov2\_vits14



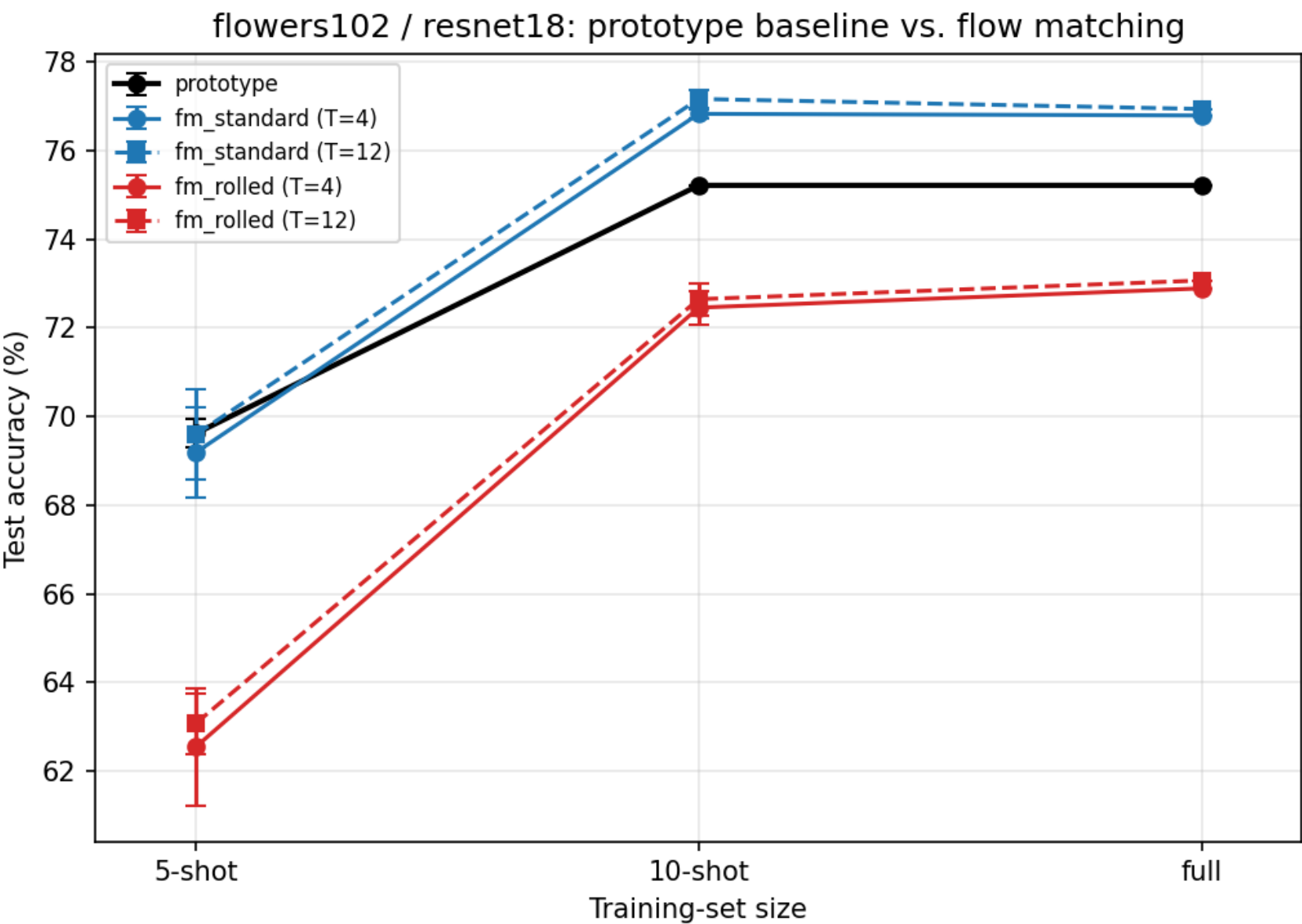
Stage 2: accuracy vs. training-set size

dtd / resnet18



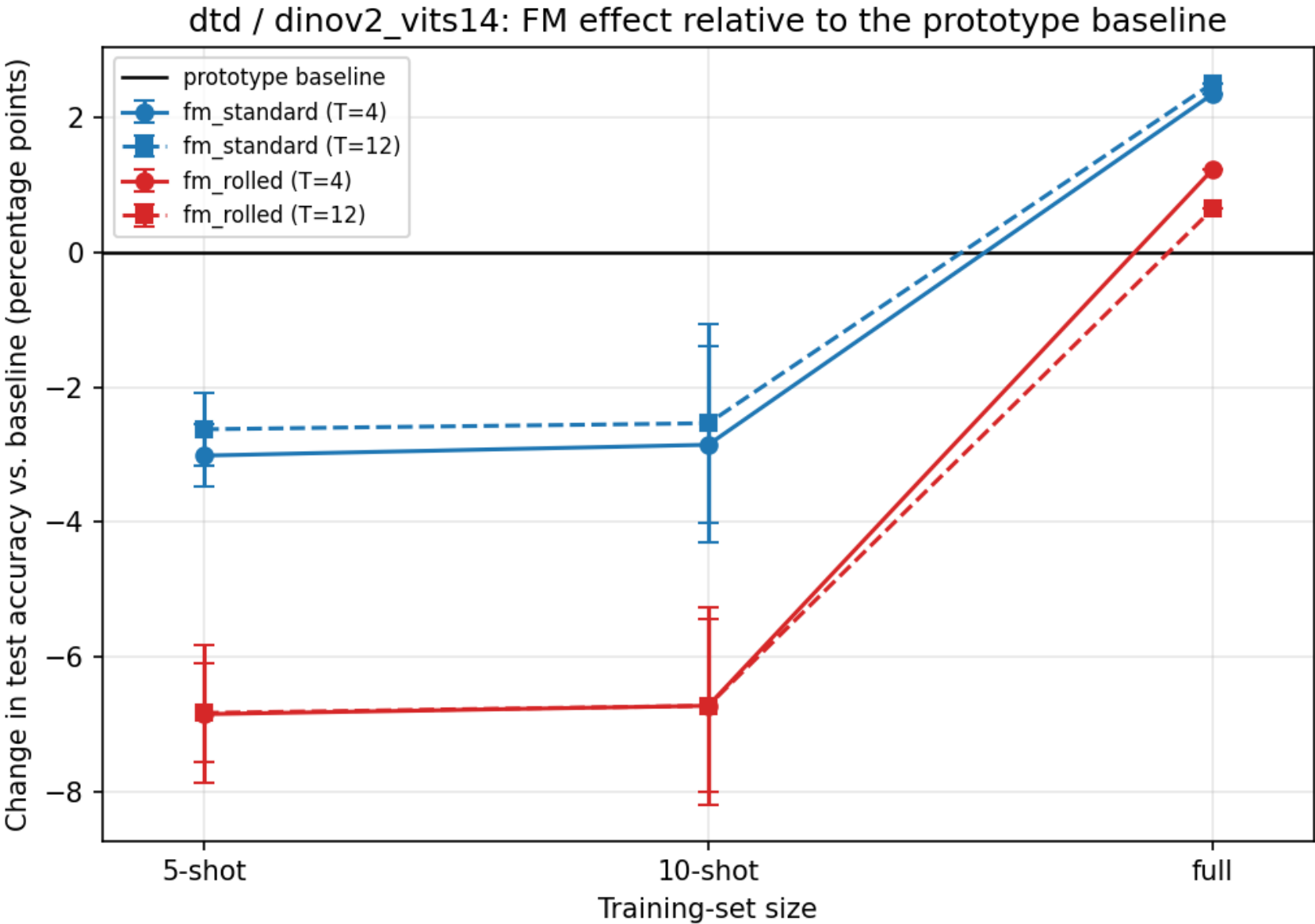
Stage 2: accuracy vs. training-set size

flowers102 / resnet18



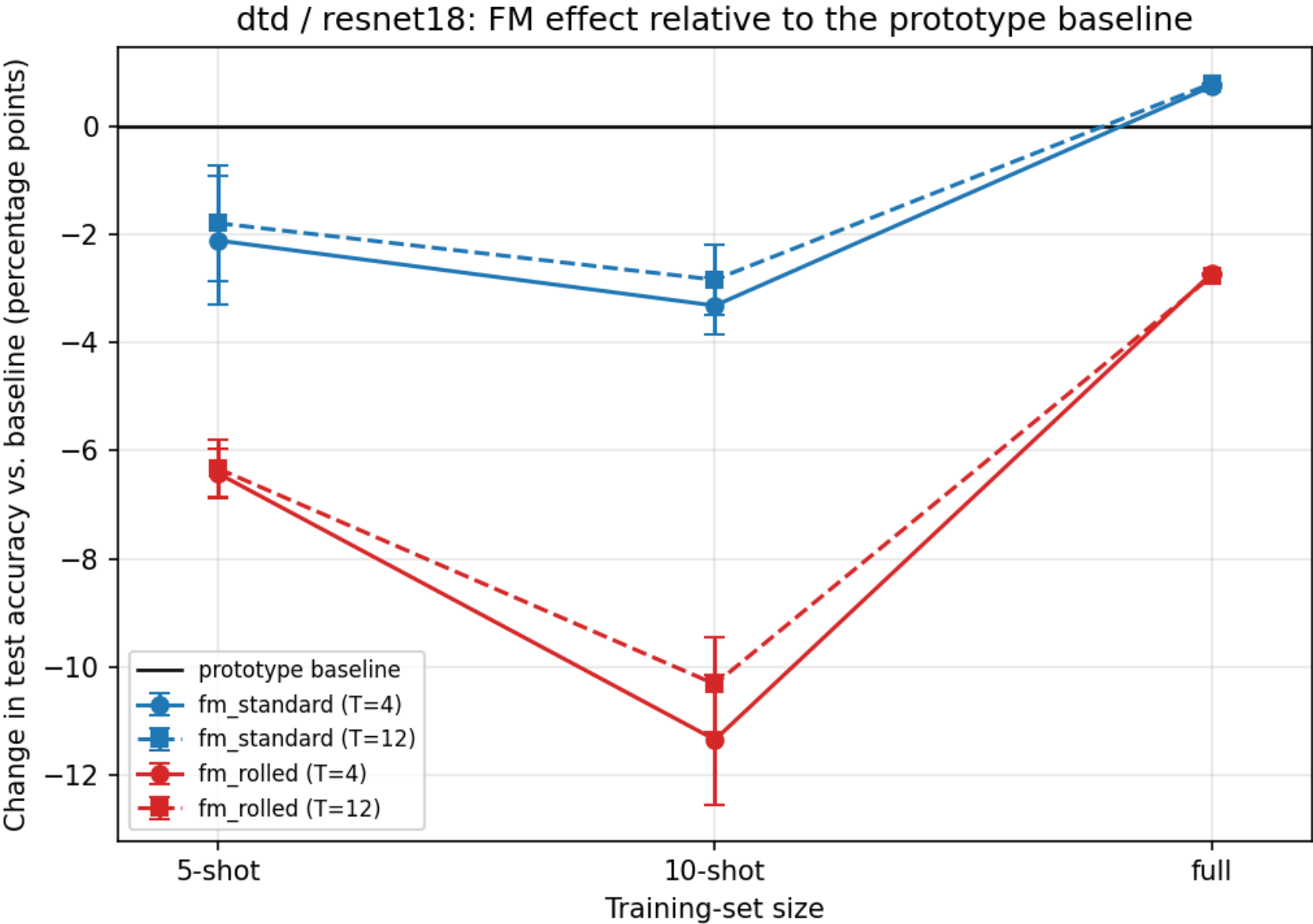
Stage 2: change in accuracy relative to the prototype baseline

dtd / dinov2\_vits14



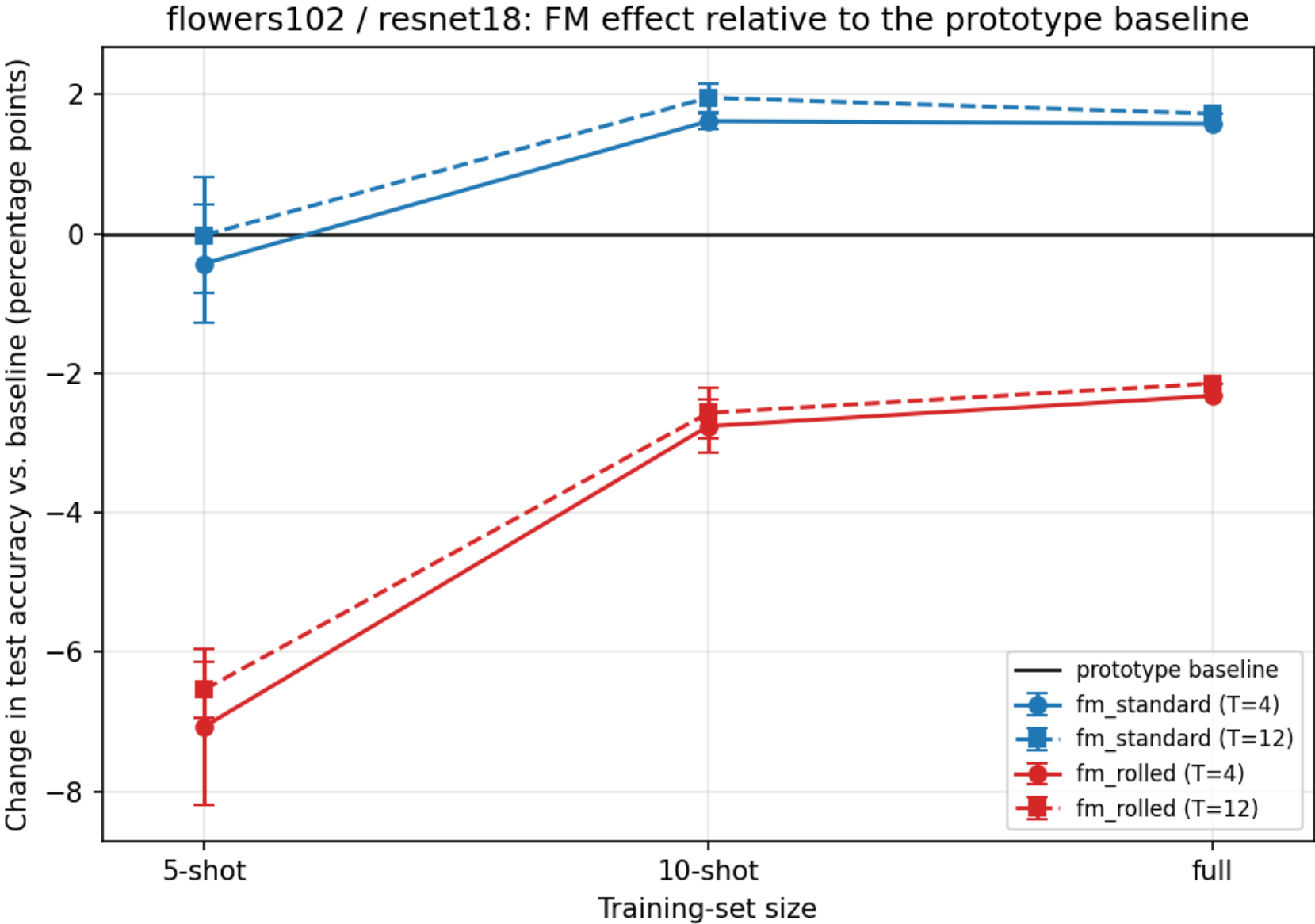
Stage 2: change in accuracy relative to the prototype baseline

dtd / resnet18



Stage 2: change in accuracy relative to the prototype baseline

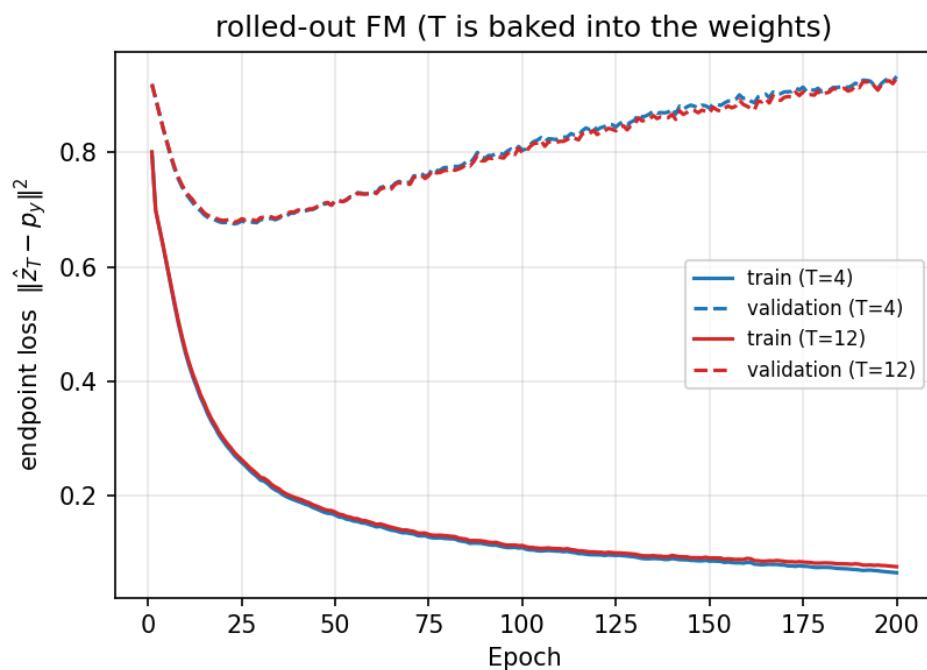
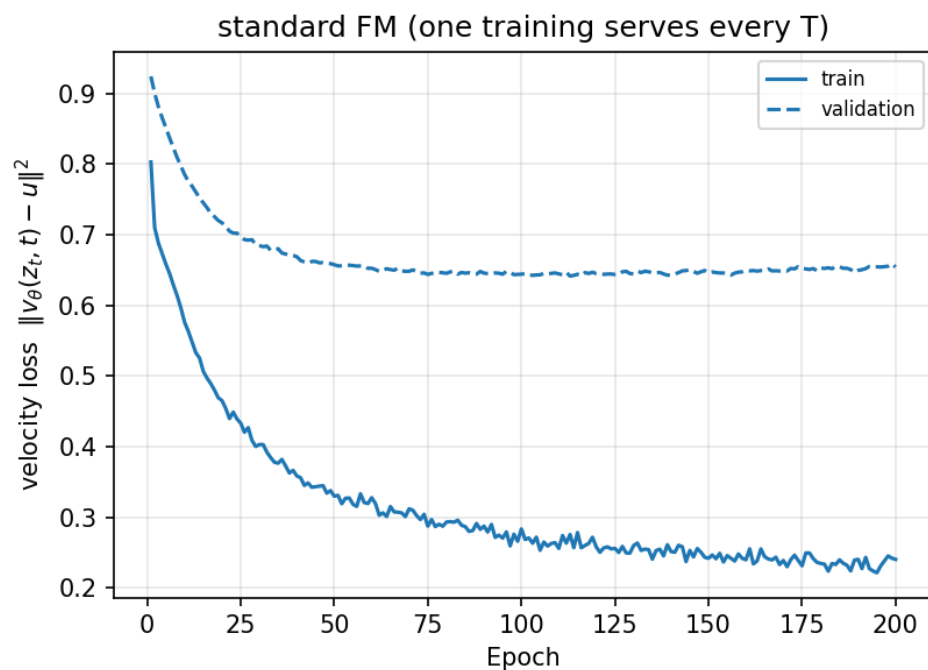
flowers102 / resnet18



## Stage 2: flow-matching training curves (10-shot, seed 0)

*dtd / dinov2\_vits14*

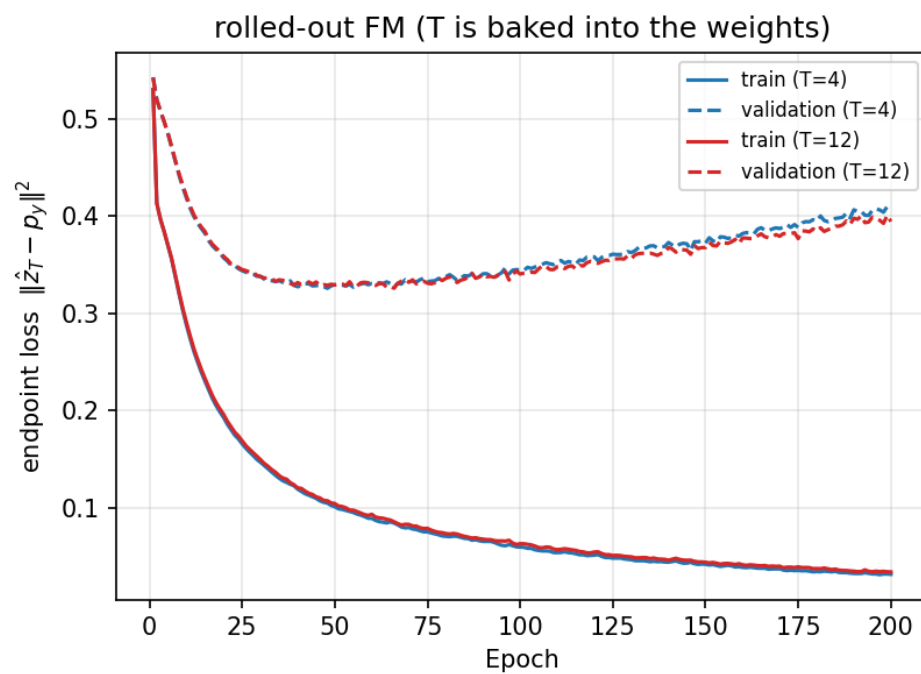
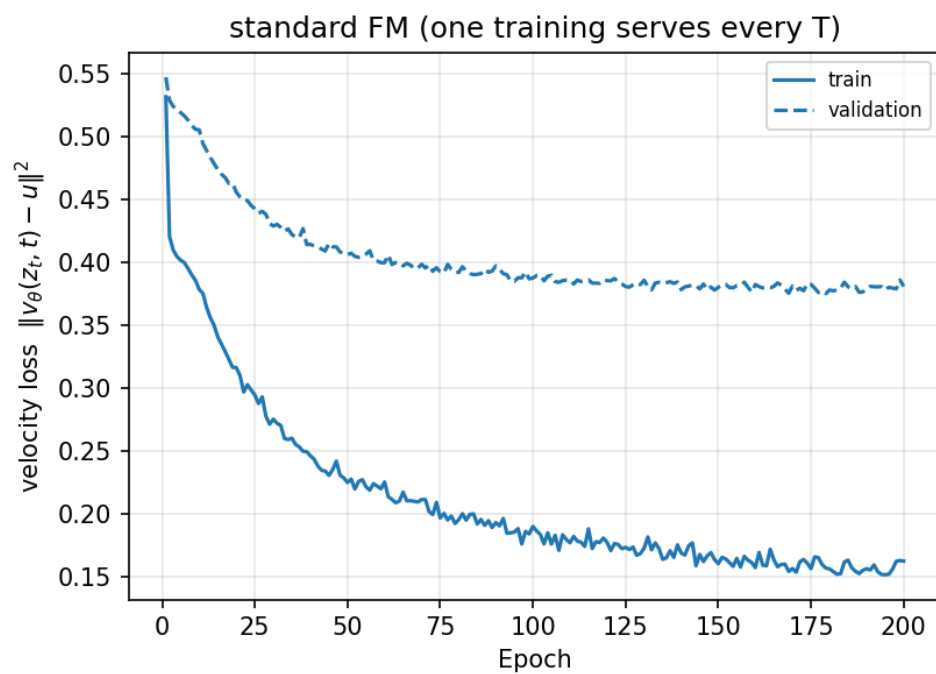
dtd / dinov2\_vits14: flow-matching training (10-shot, seed 0)  
note: the two panels measure different quantities and are not comparable to each other



## Stage 2: flow-matching training curves (10-shot, seed 0)

*dtd / resnet18*

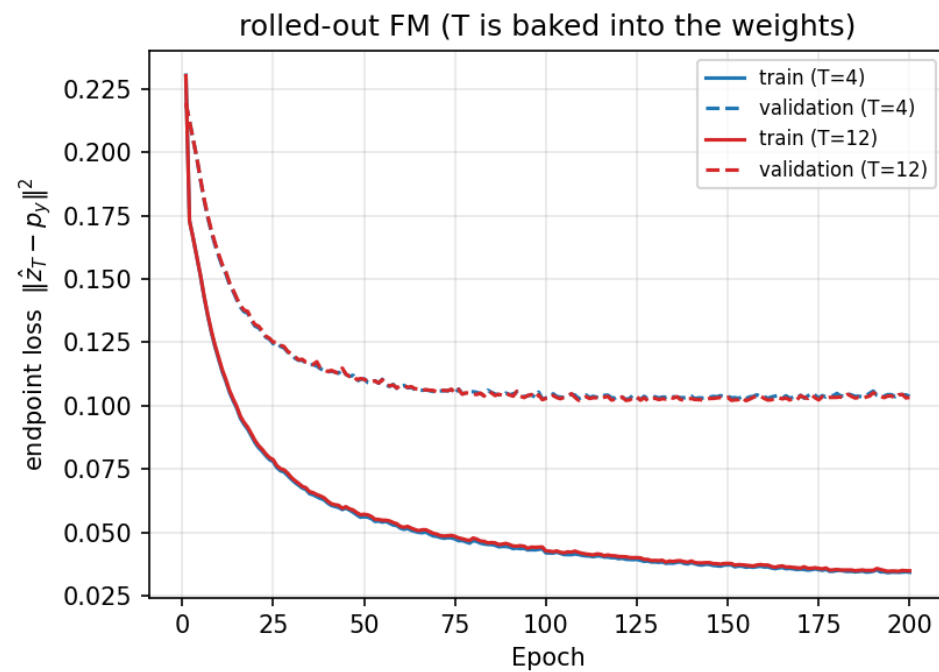
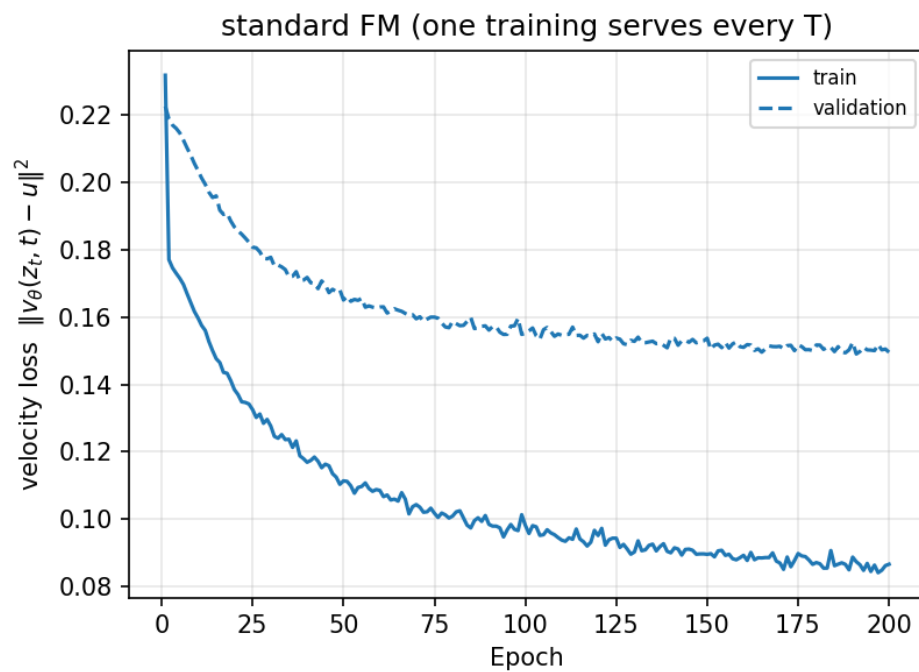
dtd / resnet18: flow-matching training (10-shot, seed 0)  
note: the two panels measure different quantities and are not comparable to each other



## Stage 2: flow-matching training curves (10-shot, seed 0)

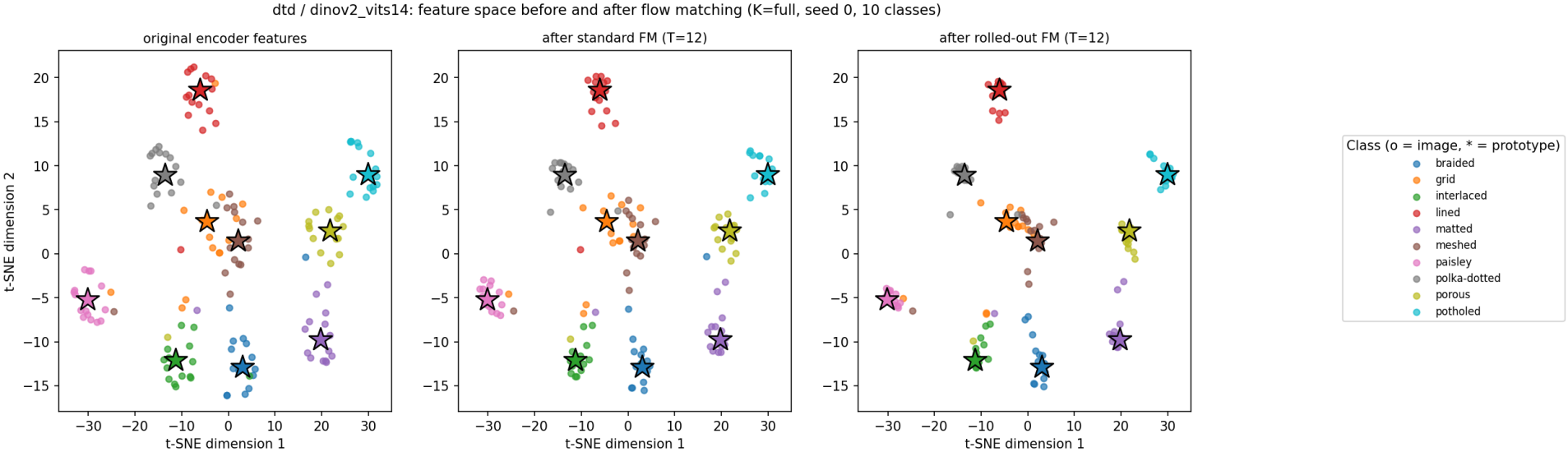
*flowers102 / resnet18*

flowers102 / resnet18: flow-matching training (10-shot, seed 0)  
note: the two panels measure different quantities and are not comparable to each other



Stage 2: feature space before and after flow matching (t-SNE)

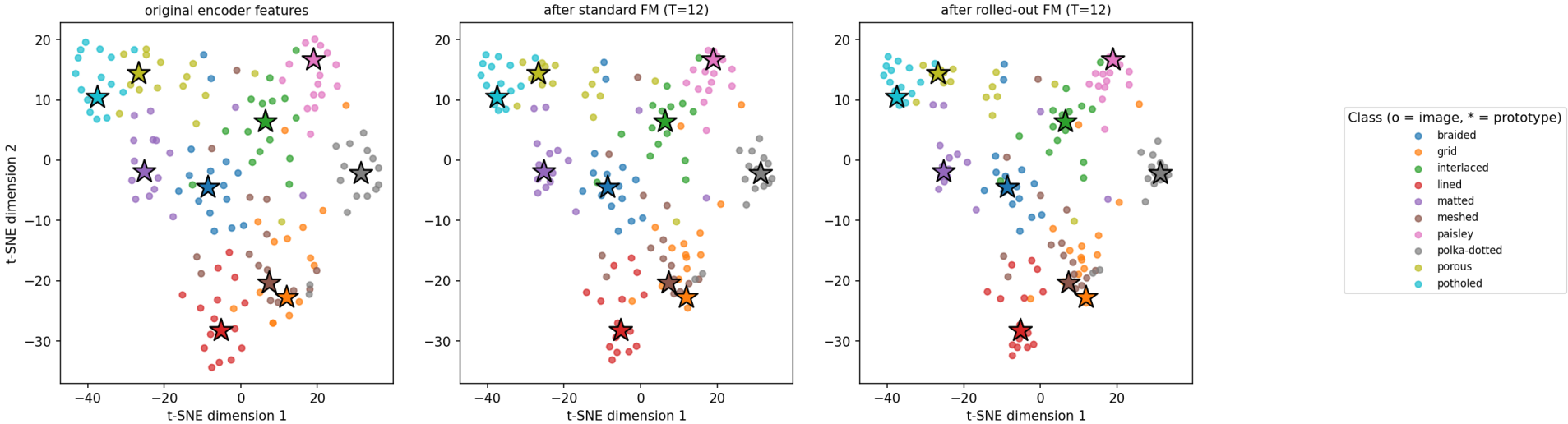
dtd / dinov2\_vits14



Stage 2: feature space before and after flow matching (t-SNE)

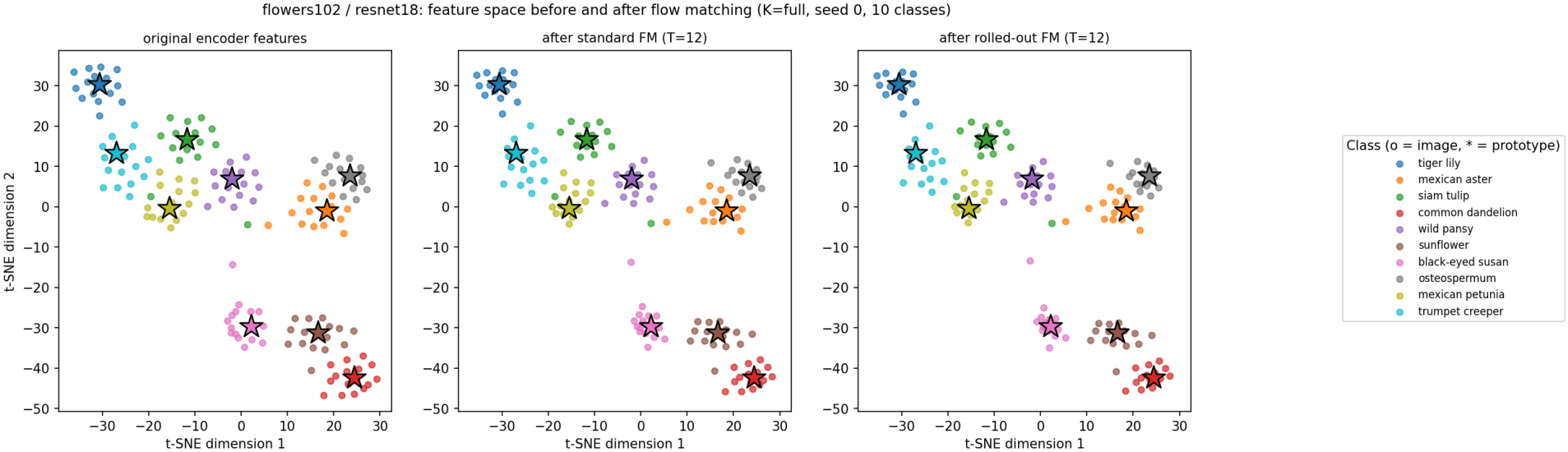
dtd / resnet18

dtd / resnet18: feature space before and after flow matching (K=full, seed 0, 10 classes)



Stage 2: feature space before and after flow matching (t-SNE)

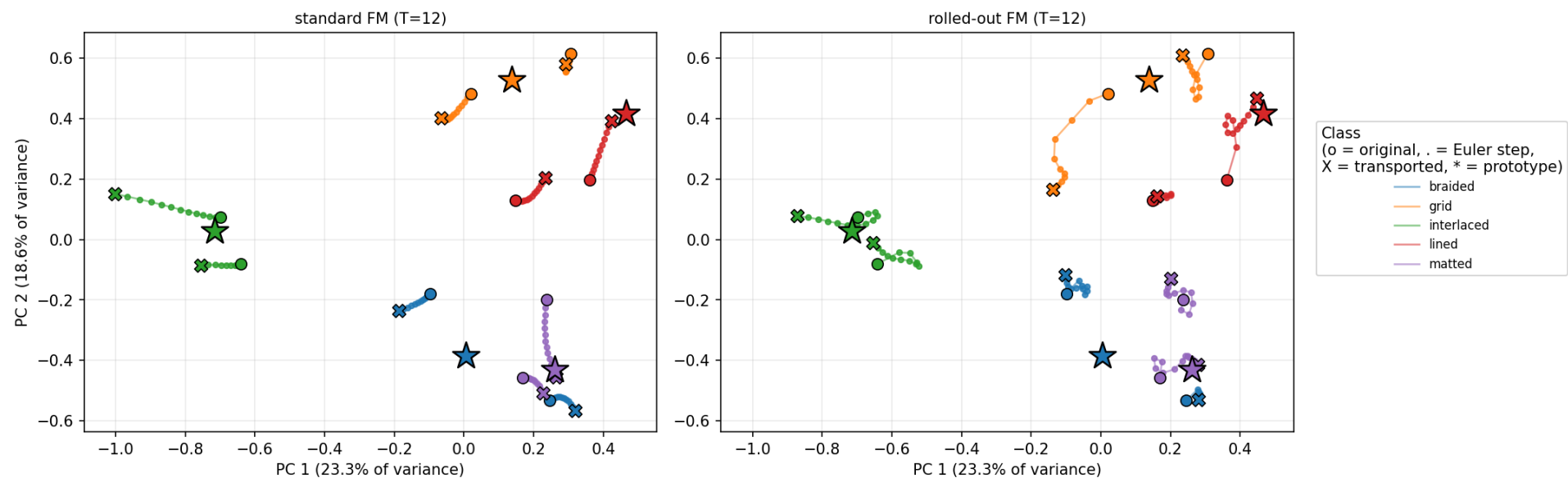
flowers102 / resnet18



Stage 2: flow trajectories (PCA)

dtd / dinov2\_vits14

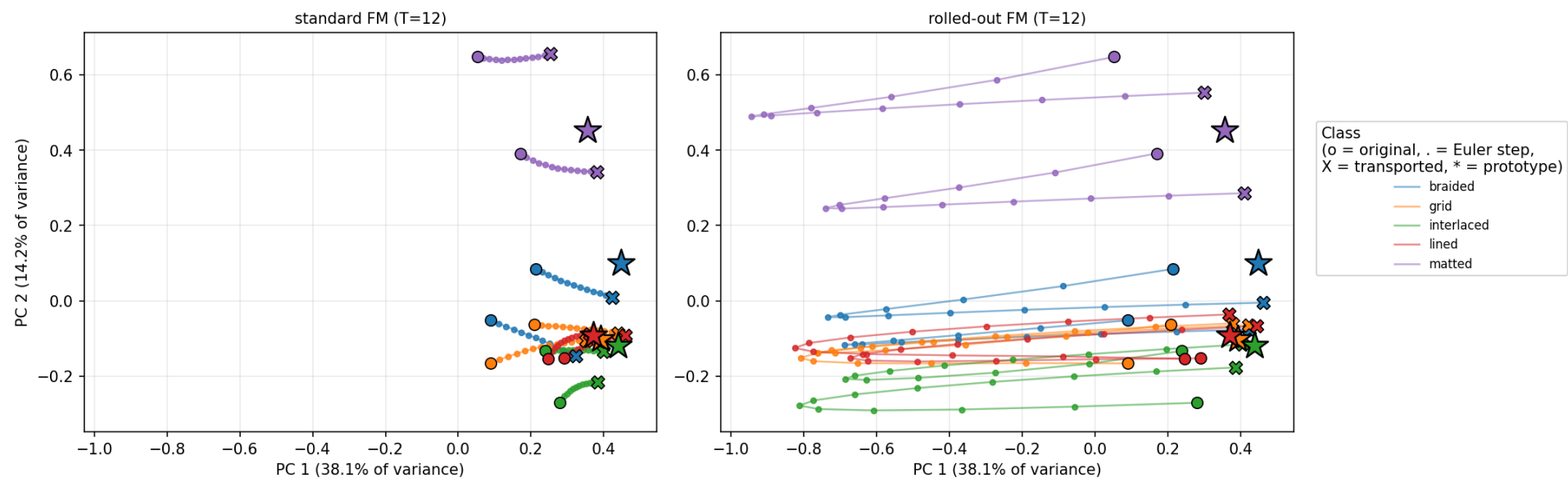
dtd / dinov2\_vits14: Euler flow trajectories (K=full, seed 0, 10 test examples)



Stage 2: flow trajectories (PCA)

dtd / resnet18

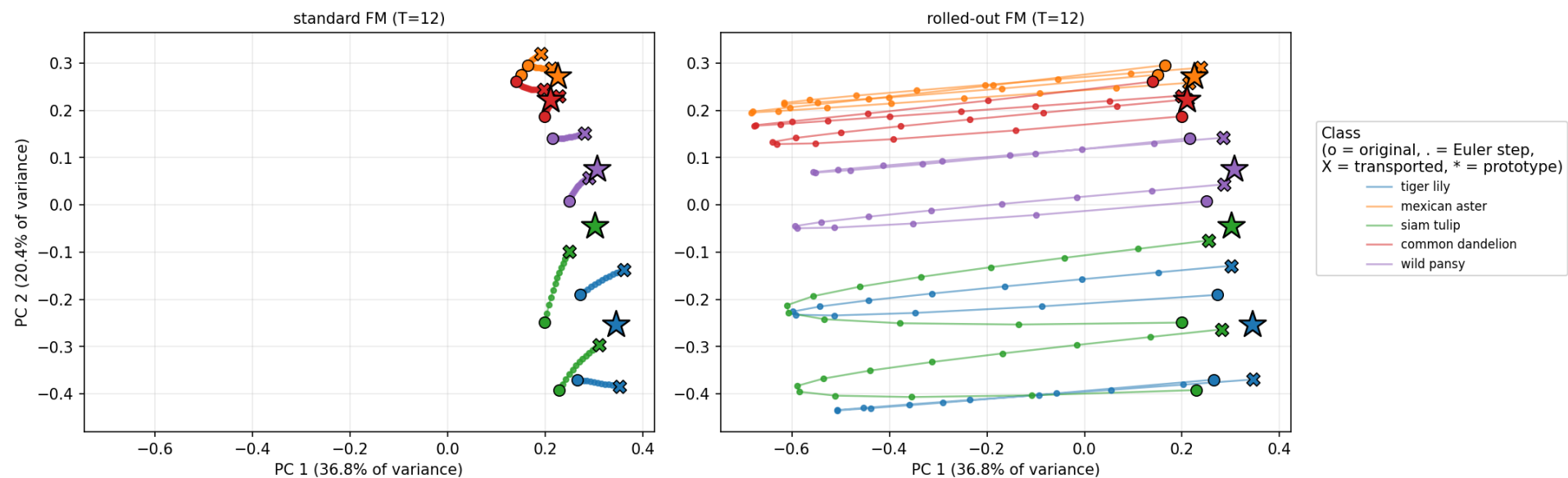
dtd / resnet18: Euler flow trajectories (K=full, seed 0, 10 test examples)



Stage 2: flow trajectories (PCA)

flowers102 / resnet18

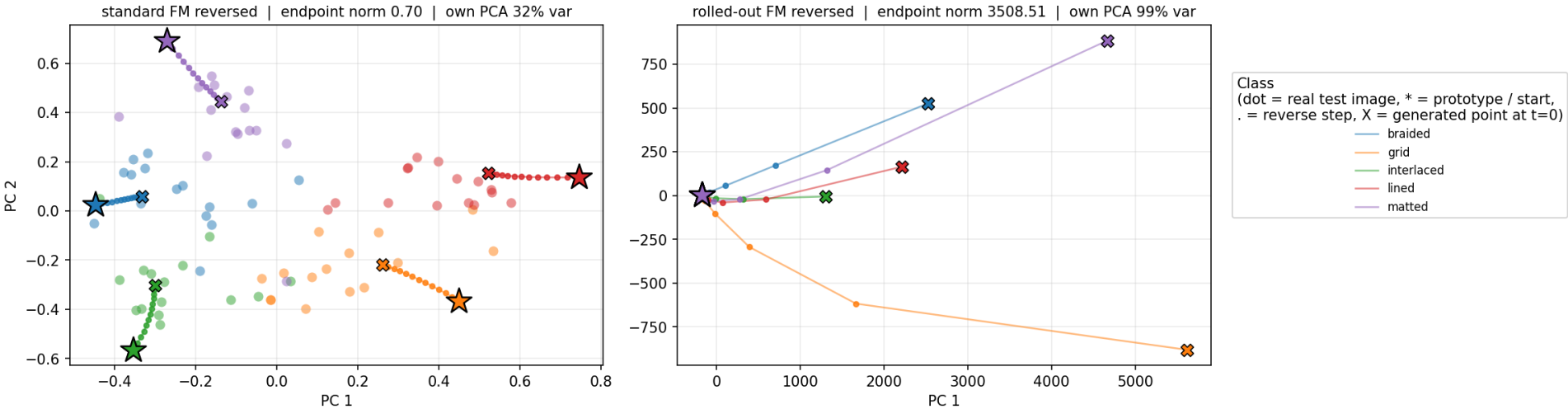
flowers102 / resnet18: Euler flow trajectories (K=full, seed 0, 10 test examples)



Stage 2: reverse flow from the class prototypes (PCA)

dtd / dinov2\_vits14

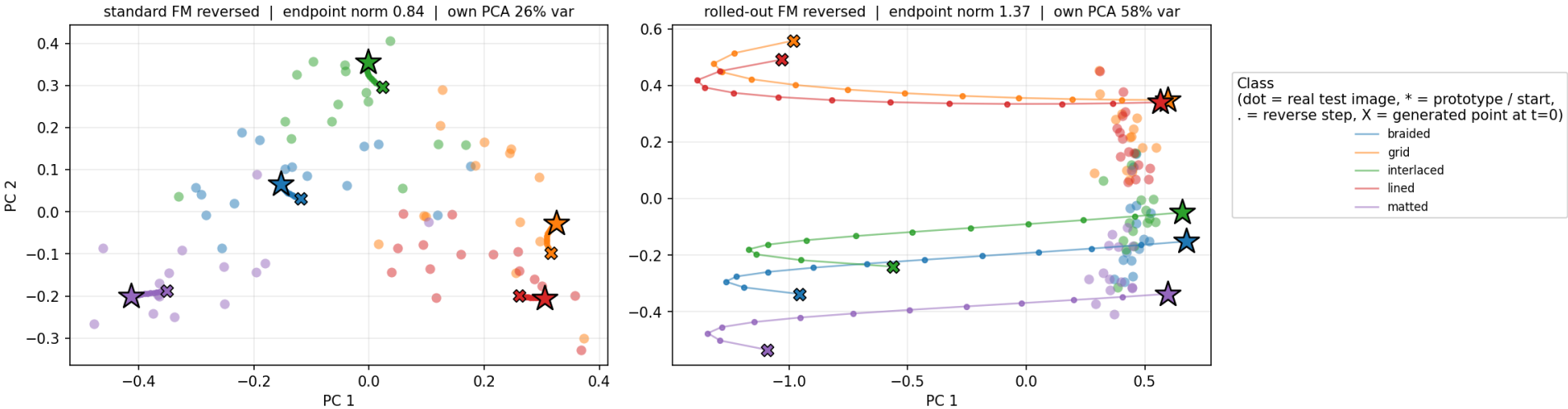
dtd / dinov2\_vits14: reverse flow from the class prototypes (K=full, seed 0, T=12)  
coloured dots are real test features of the same classes - does the generated point land among them?  
each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



Stage 2: reverse flow from the class prototypes (PCA)

dtd / resnet18

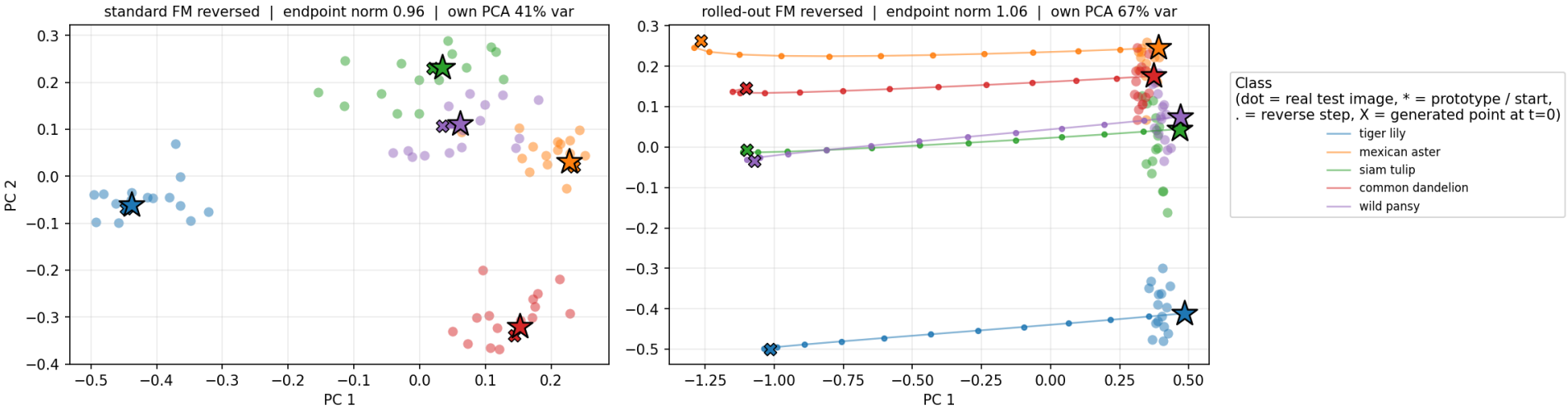
dtd / resnet18: reverse flow from the class prototypes (K=full, seed 0, T=12)  
coloured dots are real test features of the same classes - does the generated point land among them?  
each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



Stage 2: reverse flow from the class prototypes (PCA)

flowers102 / resnet18

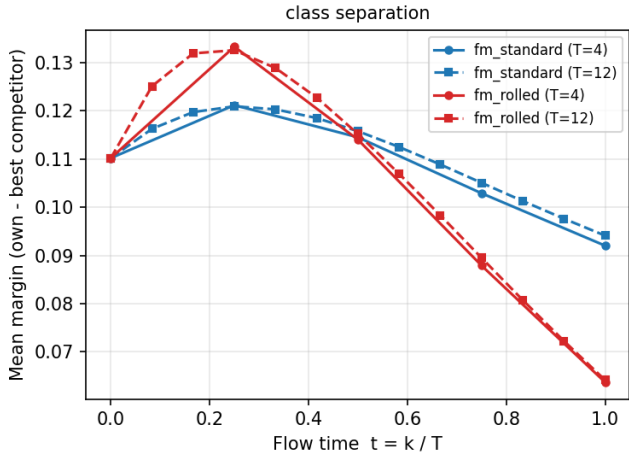
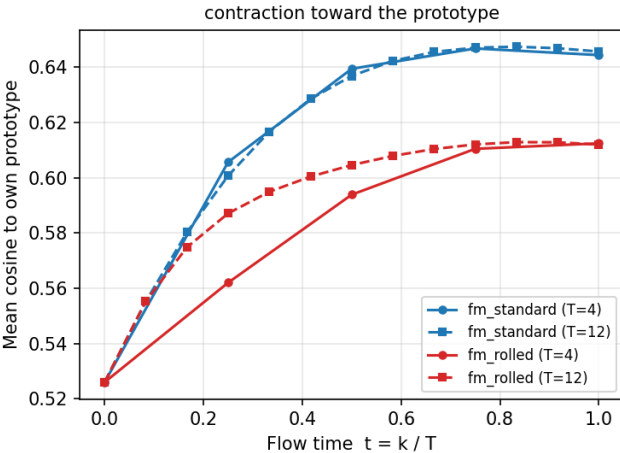
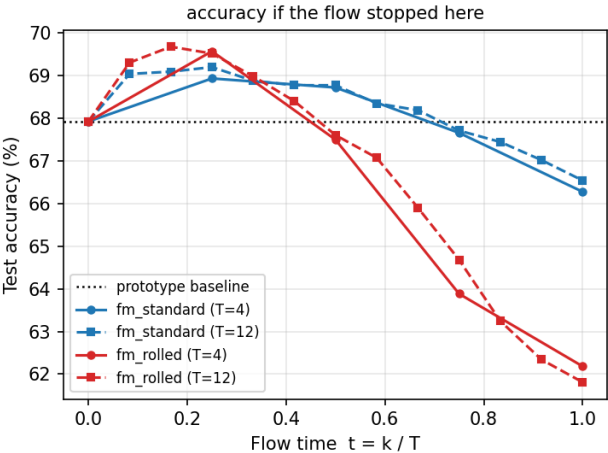
flowers102 / resnet18: reverse flow from the class prototypes (K=full, seed 0, T=12)  
coloured dots are real test features of the same classes - does the generated point land among them?  
each panel has its own PCA basis and scale - not comparable by eye; see endpoint norms



Stage 2: metrics along the flow

dtd / dinov2\_vits14 (K=10)

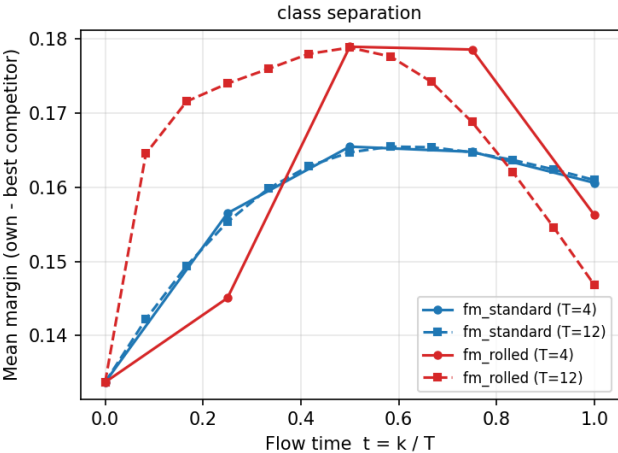
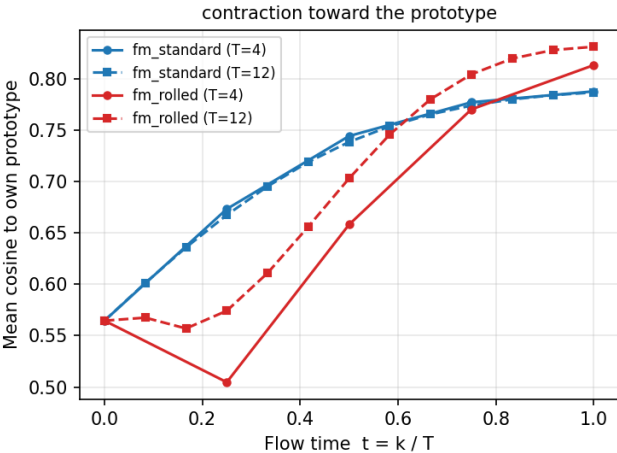
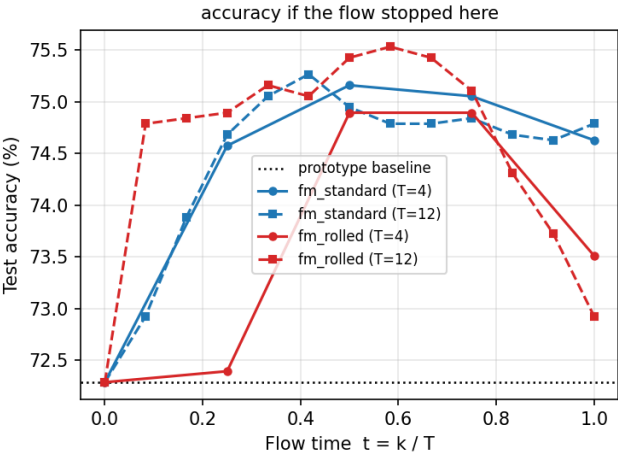
dtd / dinov2\_vits14: metrics along the flow (K=10)



Stage 2: metrics along the flow

dtd / dinov2\_vits14 (K=full)

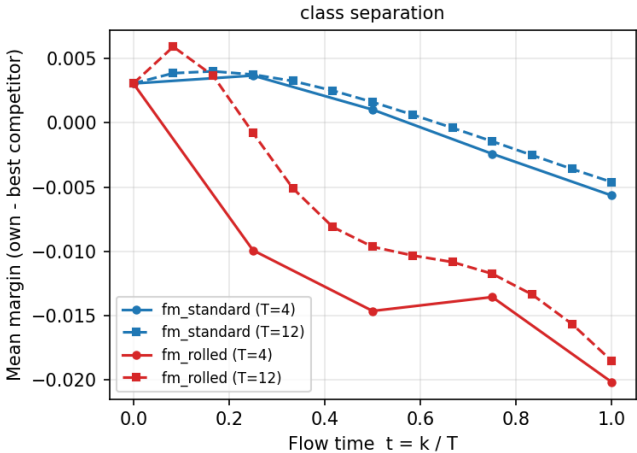
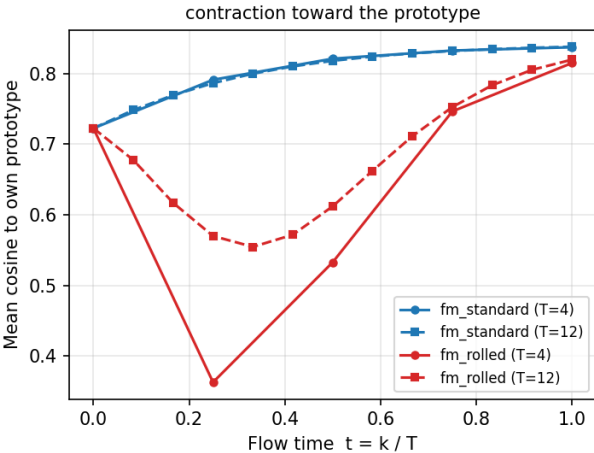
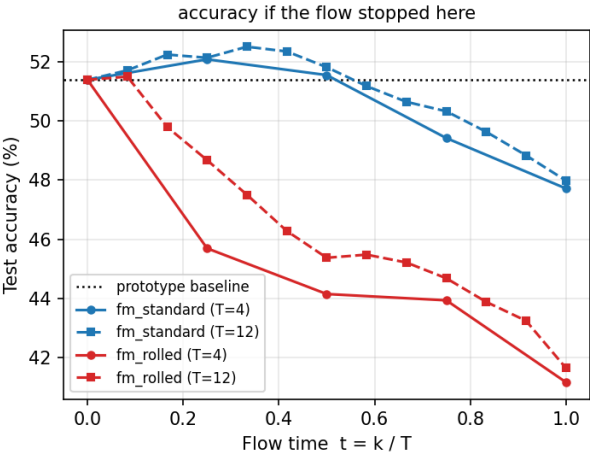
dtd / dinov2\_vits14: metrics along the flow (K=full)



Stage 2: metrics along the flow

*dtd / resnet18 (K=10)*

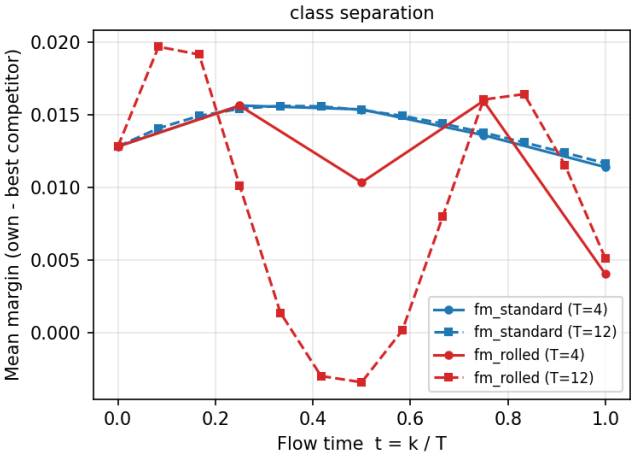
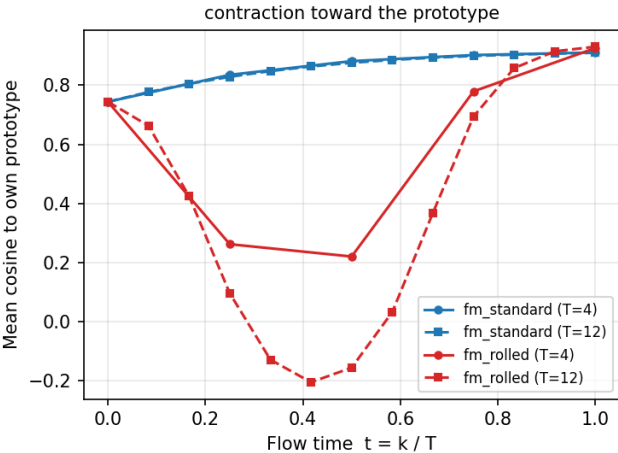
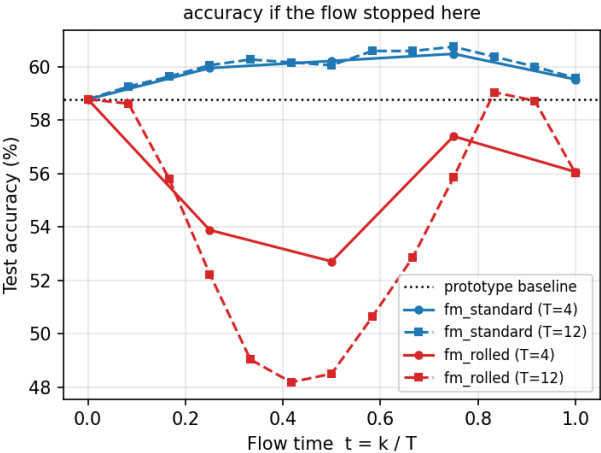
dtd / resnet18: metrics along the flow (K=10)



Stage 2: metrics along the flow

dtd / resnet18 (K=full)

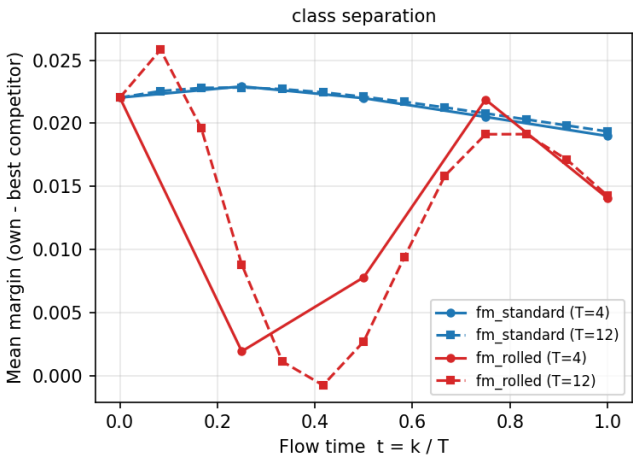
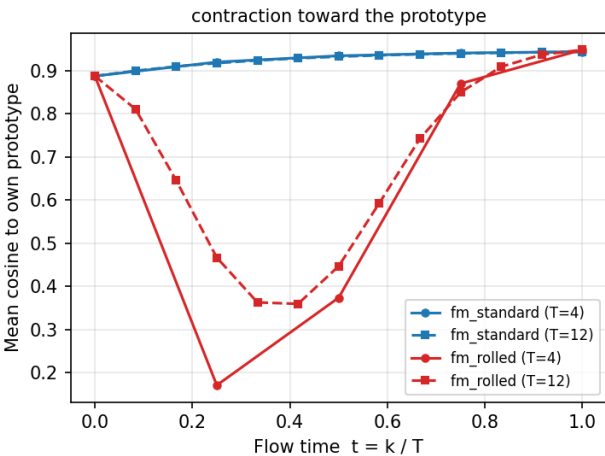
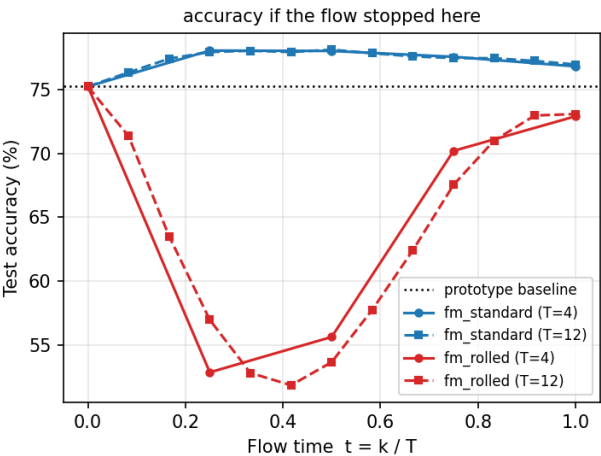
dtd / resnet18: metrics along the flow (K=full)



Stage 2: metrics along the flow

flowers102 / resnet18 (K=10)

flowers102 / resnet18: metrics along the flow (K=10)



Stage 2: metrics along the flow

flowers102 / resnet18 (K=full)

flowers102 / resnet18: metrics along the flow (K=full)

